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PLATING POSTERS INC • METAL FINISHING REFERENCE SERIES

HARDCOAT ANODIZING TYPE III

ANODIZE CLASS **TYPE III**

TYPE I

CHROMIC • THIN

TYPE II

SULFURIC • ~5-25 MM

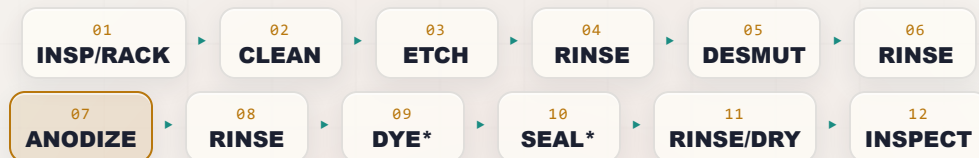
TYPE III

HARDCOAT • ~25-100+ MM

SULFURIC • HARDCOAT • WEAR/ABRASION • ~25-100+ MM • COLD BATH

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from “wait, the part is the positive electrode?” to a signed certificate of completion. The thick, hard, wear-resistant engineering anodize on aluminum. We don’t deposit metal here — we grow a dense ceramic skin out of the part. Assumes you know nothing. Teaches you everything.



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • FOR-DUMMIES STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer/aerospace specifications (e.g., MIL-A-8625 / MIL-PRF-8625 Type III, AMS 2469), and current Safety Data Sheets (SDS), and comply with OSHA, EPA, REACH, and local regulations before production use. Sulfuric acid and caustic etch cause severe burns; the hardcoat bath runs near freezing under high-current DC, and seal tanks run near boiling.

— FIND YOUR WAY

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REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, your customer’s spec (MIL-PRF-8625 Type III, AMS 2469, ASTM, etc.), or your supervisor’s instructions. When this manual and your shop’s official procedure disagree, **follow your shop’s official procedure and ask your supervisor.**

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN EXACTLY THE RIGHT PLACE — AND HERE, "ANODE" IS THE WHOLE GAME

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — a **hardcoat (Type III) anodizing line**. Whether you started this week or moved over from a Type II line, you're in the right place, and we're glad you're here.

Here is the first big thing to wrap your head around, and it's going to feel backwards if you've ever seen electroplating: **in anodizing, we are not coating your part with a different metal. We are growing a tough, glass-hard oxide layer out of the aluminum itself.** And to do that, your part is hooked up as the **anode** — the *positive* electrode — which is the exact opposite of plating. If that sentence made your eyebrows go up, good. That's the single biggest idea in this whole book, and we'll teach it from the ground up.

The second thing we'll tell you up front: **this manual assumes you know nothing about anodizing, chemistry, or electricity.** That's not an insult — it's a promise. Every term gets defined the first time we use it; every step explains not just *what* to do but *why* it matters. It's written in the spirit of the *For Dummies* books — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank of near-freezing acid running at sixty volts.



REMEMBER

You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: inspect and rack it, clean it, etch it, rinse it, desmut it, rinse it, **anodize it** (cold, high-power), rinse it, (optionally) **dye it dark**, rinse it, (optionally) **seal it**, rinse/dry it, and inspect it. Reading it in order matters, because **the order of an anodizing line is not random** — and neither is the order of this book.



TIP

Read the Fundamentals (Part One) all the way through *before* you read any individual station chapter. The station chapters assume you already understand the big picture Part One gives you — especially “the part is the anode and we grow oxide out of it” and the “cold + power = hard” idea unique to hardcoat. It's the difference between reading a recipe when you already know how to cook versus the very first time you've stepped into a kitchen.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS



TIP

A shortcut, a trick of the trade, or a smarter way to do something — the kind of thing a 20-year veteran would lean over and tell you. These save you time and headaches.



REMEMBER

A core idea worth locking in. If you forget everything else on the page, don't forget these. They're the load-bearing ideas.



HEADS-UP

A safety warning. Read every single one. These keep your skin, eyes, and lungs intact. A hardcoat line adds a near-freezing refrigeration system and *higher-voltage* DC to the usual acid, caustic, and near-boiling tanks — we use these *liberally*.



TECHNICAL STUFF

Deeper chemistry or theory for the curious. Totally optional. Skip these and you can still run the line. Read them and you'll understand it on a deeper level.



FUN FACT

A bit of interesting trivia to make the science stick (and to give you something to say at lunch).

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (sometimes paired with a red “Top Mistakes” card) — the things you must be able to *say out loud, unprompted*, before you're cleared on that station — and a bordered **Sign-Off** box your *trainer initials* once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you have to pass through.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what anodizing is** in plain language — and how it is fundamentally *different from electroplating*: the part is the **anode**, and we **grow an aluminum-oxide layer out of the aluminum itself** instead of depositing a different metal onto it.
2. **Describe what Type III hardcoat anodizing does** and why a shop chooses it — a **thick, dense, very hard, wear- and abrasion-resistant** integral oxide; the *engineering* coating, chosen for parts that must survive friction, sliding, and abuse.
3. **Explain why hardcoat is run cold and with more power** — cold suppresses the acid's dissolving action so the oxide grows **dense and hard** instead of soft and porous, while higher voltage and current density grow it **thick**.
4. **Explain dimensional growth** — the coating grows roughly **half into** the surface and **half out**, so a part gets bigger by about half the coating thickness per surface — and recognize that with a thick hardcoat **this is a large, tolerance-critical number**.
5. **Place Type III in the family** — understand how Type I (chromic), Type II (sulfuric), and Type III (hardcoat) differ in chemistry, temperature, power, thickness, and use.
6. **Explain why aluminum alloy matters even more for hardcoat** — how high-copper (2xxx) and high-silicon (cast) alloys give lower hardness, darker color, and a real risk of *burning*.
7. **Name the stations of the line in order** and explain *why* the order can't be rearranged.
8. **Recognize the main hazards** at each station — sulfuric acid, hot caustic etch, hydrogen gas at the cathode, **higher-voltage DC and greater arc/exotherm risk**, the **aggressive refrigeration system**, near-boiling seal tanks, and dye handling — and know the correct PPE and emergency response.
9. **Use the basic field checks** every anodizer relies on — water-break test, coating thickness (eddy-current and micro-section), microhardness/abrasion (Taber), and (when sealed) seal-quality tests.
10. **Spot trouble early** — especially **burning** and **cooling loss** — and **talk the language** with your lead, your chemist, and your supplier.

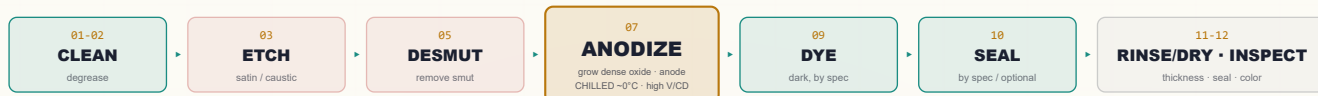


REMEMBER

You are not expected to hit all of these on day one. Hardcoat is a craft, and a demanding one. The goal is steady competence, not instant mastery — but the *safety* objectives (#8) are not optional and not gradual. You must have those before you work the line at all.

• THE LINE AT A GLANCE

The whole Type III line in one strip. Don't memorize yet — feel the rhythm: **Inspect/Rack** → **Clean** → **Etch** → **Rinse** → **Desmut** → **Rinse** → **ANODIZE** (cold, high-power) → **Rinse** → **(Dye, dark only)** → **Rinse** → **(Seal, by spec)** → **Rinse/Dry** → **Inspect**.



REMEMBER

Notice the pattern: **a rinse follows almost every working tank**. Rinses aren't breaks — they're checkpoints that keep one chemistry from poisoning the next. On a hardcoat line a bad rinse before the anodize tank can quietly wreck an expensive, long-running job. They aren't filler; they're some of the most important steps you'll perform.



FUN FACT

The aluminum oxide we grow in this tank is chemically the same family as **sapphire and ruby** (crystalline aluminum oxide, Al_2O_3). On hardcoat we grow it so dense that the finished surface can approach the hardness of those gems — you're not bolting on a coating, you're turning the skin of the part into engineering ceramic.

CHAPTER 1

WHAT IS ANODIZING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT ANODIZING IS

THE ONE-SENTENCE VERSION

Anodizing is using electricity to grow a hard, protective oxide layer out of the surface of an aluminum part — the part itself becomes part of the coating. That's the whole idea. Now let's unpack it, because it's genuinely different from plating and that difference trips up almost everyone at first.

FIRST, WHAT PLATING DOES (SO YOU CAN SEE THE DIFFERENCE)

In electroplating — zinc, nickel, chrome — you take your part, hook it up as the **cathode** (the *negative* electrode), and you *deposit a different metal* onto it from a bath. Zinc goes onto steel. Chrome goes onto steel. The part collects metal that came from somewhere else (dissolving anodes or the bath chemistry). The coating is a *separate metal sitting on top* of your part.

Anodizing flips almost all of that:

- Your part is the **anode** — the **positive** electrode. (That's literally where the word "anodize" comes from.)
- We don't add a different metal. We take the **aluminum that's already there** and convert its surface into **aluminum oxide** by feeding electricity through an acid bath.
- The coating isn't sitting *on* the part — it **is** the part's surface, chemically grown and locked in. It can't peel like paint or flake like a bad plate, because it's integral to the metal.



REMEMBER — THE SINGLE BIGGEST IDEA IN THIS BOOK

In plating, the part is the **cathode** and you **add** a metal. In anodizing, the part is the **anode** and you **grow oxide out of the part itself**. If you remember nothing else from Chapter 1, remember this. Everything else follows from it.

HOW IT PHYSICALLY WORKS: THE KITCHEN-SINK PICTURE

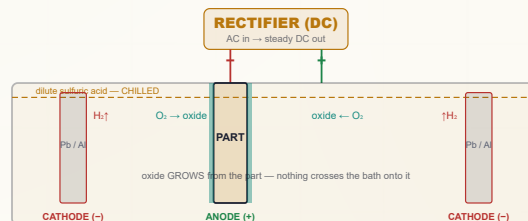
Picture a tank full of liquid called the **electrolyte** (or just "the bath"). For hardcoat, that liquid is **dilute sulfuric acid** — sulfuric acid mixed with water, very much like Type II — but here it is **kept extremely cold**. You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction "DC" current anodizing needs):

- **The anode** — this is your **aluminum part**, connected to the **positive** terminal. "*Anode = the part, the thing growing oxide.*"
- **The cathode** — bars or plates (commonly **lead** or **aluminum 6063**) hanging in the same tank, connected to the **negative** terminal. The cathode just completes the circuit and is where **hydrogen gas** bubbles off.

When you turn on the rectifier, here's the dance that happens:

1. Current flows. At the surface of your aluminum part (the anode), water in the bath is split and **oxygen** is delivered right at the metal surface.
2. That oxygen immediately combines with the surface aluminum to form **aluminum oxide (Al₂O₃)** — a hard, ceramic layer.
3. The sulfuric acid is slightly aggressive, so as the oxide forms, the acid is *also* dissolving it — a tug-of-war. **In hardcoat we deliberately keep the bath very cold so that the acid barely dissolves the oxide.** That tips the tug-of-war strongly toward *building*, so the oxide grows **thick, dense, and hard** with much smaller, tighter pores than warm Type II makes.
4. Over at the cathode, **hydrogen gas** bubbles off harmlessly into the air (this is a safety item — flammable gas — we'll come back to it).

The net result: the surface of your part *converts* into a thick, dense, very hard oxide layer that grows thicker the longer you anodize and the more current you push — and grows *harder* the colder you keep it.



Compare to a plating cell: the labels are reversed — here the **part is the ANODE (+)** and the counter-electrodes are the cathodes.



TECHNICAL STUFF — THE TWO REACTIONS

At the part (anode), aluminum is oxidized: simplified, $2\text{Al} + 3\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 6\text{H}^+ + 6\text{e}^-$. At the cathode, those electrons reduce hydrogen ions: $6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2\uparrow$. So **oxygen builds the coating at your part; hydrogen gas leaves at the counter-electrode**. No aluminum travels through the bath onto something else — it stays put and turns to oxide in place.



TECHNICAL STUFF — WHY HARDCOAT IS RUN NEAR FREEZING

The reaction is **exothermic** (it makes heat), and on hardcoat we push *much more* electrical power than Type II, so *much more* heat is dumped right at the part surface. Two things follow. First, cold suppresses the acid's dissolving action — so the coating grows **dense and hard** instead of soft and porous; this is the entire point of hardcoat. Second, because we're making so much more heat, we need a **powerful refrigeration system** to hold the bath near $\sim 0^\circ\text{C}$ ($\sim 32^\circ\text{F}$). On hardcoat, **cold and well-stirred is the whole product** — and if the cooling can't keep up, the part *burns*.



FUN FACT

Anodizing is one of the few finishing processes where the coating is *grown from* the part rather than *added to* it. That's why hardcoat aluminum can't "chip off" to bare metal the way paint or a bad plate can — there's no glue line. The oxide and the metal are continuous, which is exactly why a thick hardcoat can take a beating.

CHAPTER 2

WHAT IS “TYPE III”?

THE HEAVY-DUTY ENGINEERING ANODIZE — THICK, DENSE, AND BUILT TO SURVIVE WEAR, FRICTION, AND ABUSE

Type III anodizing — almost always called “**hardcoat**” (or “hard anodize”) — is a **sulfuric-acid anodize run cold and at high power** to grow a **thick, dense, exceptionally hard, wear-resistant** aluminum-oxide layer, typically **~25–100+ µm (0.001–0.004”+)**, with **~50 µm (0.002”)** a very common target. The “Type III” name comes from **MIL-A-8625 / MIL-PRF-8625** (Type I chromic, Type II sulfuric, Type III hardcoat); the aerospace spec **AMS 2469** also covers hardcoat.

The key idea: Type III uses the **same family of sulfuric chemistry as Type II**, but the **process** is different — *colder bath, higher voltage (ramped up), higher current density, longer time, thicker and harder result*. It is chosen when a part has to *do a job mechanically*, not just look good or resist weather.

• WHY SHOPS CHOOSE IT

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Extreme hardness	The oxide itself is very hard (microhardness often cited near the high end of the Rockwell C range, ~60–70 Rc-equivalent). It resists scratching, galling, and wear far beyond Type II.
Wear / abrasion resistance	Hardcoat’s headline. Sliding parts, cylinders, pistons, gears, cams, and guides last far longer.
Thick & dense	Typically ~25–100+ µm — many times thicker than Type II — and much denser because of the cold bath.
Corrosion resistance	Excellent, especially if sealed (but see the wear-vs-seal tradeoff in Ch 3 and Station 10).
Electrically insulating	A thick, dense oxide is an even better insulator than Type II — useful for dielectric standoff.
Integral & non-flaking	Because it’s grown from the metal, it won’t peel or chip to bare aluminum.
Base for dry-film lube / PTFE	The dense-but-porous surface holds impregnated PTFE or other lubricants for low-friction wear surfaces.

WHERE YOU’LL SEE IT

Hydraulic and pneumatic cylinders and pistons, valve bodies, gears, cams, sliding mechanisms, pump components, military and aerospace structural and wear parts, firearms components, molds and tooling, sheaves and pulleys, robotics wear surfaces, and high-end gear that must survive hard use. Where Type II is “decorative + protective,” **Type III is “engineering + wear.”**

★

REMEMBER

Type III’s superpower is **hardness and wear resistance from a thick, dense oxide**. That comes from running the *same sulfuric chemistry* as Type II but **colder and with more electrical power**. Color and decoration are usually *not* the point — the coating’s natural color is a gray/bronze/dark that varies with the alloy, and it’s typically left that way or dyed only dark colors.

💡

TIP

When a print says “**hardcoat**,” “**hard anodize**,” “**Type III**,” “**MIL-A-8625 Type III**,” or “**AMS 2469**,” that’s *this* process — not ordinary Type II. Chemistry, temperature, voltage, thickness, color, and dimensional growth all change. If a print just says “anodize,” it usually means **Type II** — check the spec and ask before you run it on the hardcoat line.

☀️

FUN FACT

A well-run hardcoat can be so hard and dense that it will **dull a file** and resist scratching from hardened steel. That’s the same Al₂O₃ family as sapphire — and it’s why hardcoat is the go-to finish for parts that have to rub against other metal for years without wearing out.

CHAPTER 3

THE POROUS OXIDE

THE STRANGEST, MOST IMPORTANT IDEA ON THE LINE — AND THE REASON COLD MAKES HARDCOAT HARD

This is the chapter that makes hardcoat *hardcoat*. Slow down here.

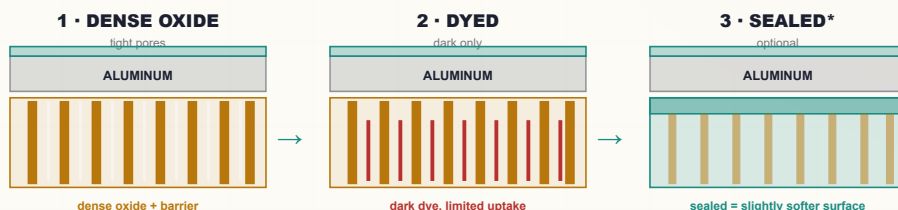
THE OXIDE STILL GROWS WITH PORES — BUT TIGHT, DENSE ONES

Remember the tug-of-war from Chapter 1: oxide forms at the surface while the sulfuric acid gently dissolves it. The result is a layer of **billions of tiny columns with a microscopic pore down the center of each one**, like a honeycomb of straws standing up on the metal, over a thin dense **barrier layer** against the aluminum. So far that's identical to Type II.

Here's the hardcoat difference: because we keep the bath **very cold**, the acid barely dissolves the growing oxide, so the pore walls stay **thick and the pores stay small and tight**. The coating ends up **much denser and harder** than the open, porous oxide of a warm Type II bath. Same architecture — *much sturdier construction*.

★ REMEMBER — COLD MAKES HARD

Warm bath → the acid eats the oxide as fast as it forms → a **soft, open, porous** coating (good for dyeing — that's Type II). Cold bath → the acid hardly touches the oxide → a **dense, hard, thick** coating (good for wear — that's Type III). The temperature of the tank is the single biggest lever between “decorative anodize” and “engineering hardcoat.”



CAN YOU DYE IT? (SOMETIMES — DARK ONLY)

Because the pores are tighter and the coating's natural color is already a gray/bronze/dark, hardcoat is **usually left its natural color**, or dyed only **dark colors (most often black)**. Bright reds and blues that pop on Type II come out muddy on dense hardcoat. Dyeing is optional and limited (Station 09).

SHOULD YOU SEAL IT? (THE BIG TRADEOFF)

On Type II you almost always seal. **On hardcoat, sealing is a genuine tradeoff:** sealing **improves corrosion resistance** but **slightly reduces hardness and abrasion resistance** (the boehmite at the pore mouths is softer than as-anodized oxide). The choice follows the part's job: **max wear → often unsealed; corrosion → sealed**. MIL-A-8625: **Class 1 = unsealed, Class 2 = sealed (and/or dyed)**. Your print tells you which.

★ REMEMBER — THE SEAL-VS-WEAR TRADEOFF (THE SOUL OF HARDCOAT DECISIONS)

Seal = better corrosion, slightly less hard. Unsealed = maximum hardness/wear, less corrosion protection. There is no “always seal” rule on hardcoat the way there is on Type II. **Follow the print: Class 1 = unsealed (wear), Class 2 = sealed (corrosion/dyed).** When in doubt, ask — you can't un-seal a part.

⚠ HEADS-UP — SEALING IS STILL A ONE-WAY DOOR

Just like Type II: once you seal, you **cannot dye** anymore, and you can't “un-seal.” If a part is to be both dyed and sealed, the order is non-negotiable: **dye before seal, never after**. And if the print calls for *unsealed* Class 1 hardcoat, **do not seal it** — you'd be quietly throwing away the hardness the customer paid for.

⚙ TECHNICAL STUFF — WHY HOT-WATER SEALING SLIGHTLY SOFTENS HARDCOAT

Near-boiling water converts the oxide at the pore walls into **boehmite** (hydrated aluminum oxide, $\text{Al}(\text{OH})_3$) that swells and closes the pore mouths. That swollen hydrated layer is excellent for corrosion resistance but is **mechanically softer** than the dense as-anodized Al_2O_3 . On thin Type II nobody cares; on a wear-critical hardcoat that small surface softening can measurably lower abrasion resistance — which is exactly why max-wear hardcoat is often left unsealed.

CHAPTER 4

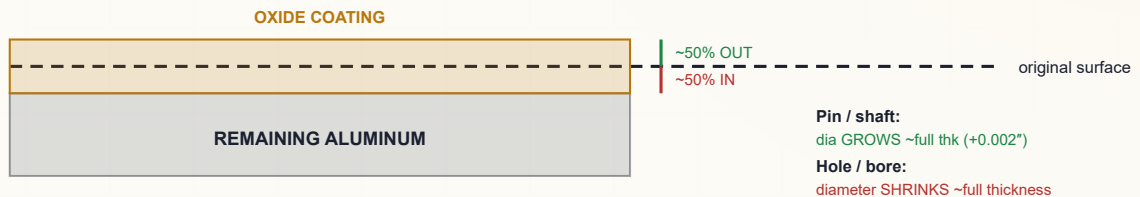
DIMENSIONAL GROWTH

THE COATING MAKES THE PART BIGGER — AND ON HARDCOAT THAT'S A NUMBER YOU ABSOLUTELY MUST PLAN FOR

Because we're converting aluminum into aluminum oxide, and oxide takes up **more room** than the metal it came from, the coating grows in **two directions at once**:

- About **half of the coating thickness grows into** the original surface (consuming aluminum).
- About **half grows out** from the original surface (the part gets bigger).

This is the same physics as Type II — **but on hardcoat the numbers are much larger**, because the coating is much thicker. For a typical **50 μm (0.002")** hardcoat, the surface moves outward by roughly **25 μm (0.001") per face**. On a shaft or pin, the **diameter grows by about the full coating thickness** (~50 μm / 0.002" on diameter); a hole's **diameter shrinks by about the full thickness**. On a thick 100 μm (0.004") hardcoat those numbers double again. That is enough to seize a bearing fit, bind a piston, or close a bore out of tolerance if nobody planned for it — **on hardcoat, dimensional growth is a primary engineering concern, not a footnote**.



REMEMBER — THE 50% RULE, WITH BIGGER STAKES

Hardcoat grows **roughly half in, half out**. Each *surface* moves outward by about **half** the coating thickness; a *diameter* (two faces) changes by about the **full** coating thickness. Because hardcoat is thick (~25–100+ μm), this is a **large, tolerance-critical** number. Tight-fit parts must be **machined undersize/oversize on purpose** or **masked** before anodizing.



TIP

When a machinist asks “how much do I undersize for hardcoat?”, the rule of thumb is: allow about **half the spec'd coating thickness per surface** (so the *full* thickness on a diameter). For a 0.002" hardcoat that's roughly **0.001" off each surface / 0.002" on diameter**. Threads, bearing fits, and press-fits matter a lot. Always confirm with the print — many hardcoat drawings call out **pre-anodize and post-anodize dimensions separately**.



HEADS-UP — MASKING AND THREADS MATTER MORE ON HARDCOAT

Areas that must *not* grow (threaded holes, bearing fits, electrical-contact points, grounding surfaces, precision bores) are usually **masked off** — and hardcoat masking is **tougher duty** than Type II masking (colder, more aggressive bath, longer run). Use masking rated for hardcoat. Anodize is also **electrically insulating** (even more so when thick), so any surface that must conduct (a ground path) *must* be masked or it won't work after anodizing.



TECHNICAL STUFF — THE PILLING-BEDWORTH IDEA

Aluminum oxide occupies more volume than the aluminum it formed from (the “Pilling–Bedworth ratio” for Al_2O_3 is greater than 1). That volume expansion is *why* the coating grows outward at all, and it's the same expansion that lets hot-water sealing pinch the pores shut. One simple fact — oxide is bulkier than metal — explains both dimensional growth *and* sealing. On hardcoat the coating is just thick enough that the effect becomes a real machining problem.



FUN FACT

Because hardcoat dimensional growth is so predictable (~half in, half out), some precision shops **deliberately anodize a part slightly small and let the coating grow it to final size** — using the oxide itself as a controlled dimension. It's one of the rare coatings precise enough to “machine with chemistry.”

CHAPTER 5

THE TYPE FAMILY

SAME IDEA, THREE FLAVORS — AND THIS MANUAL IS THE HEAVY-DUTY ONE

All three “types” grow an integral aluminum-oxide coating with the part as the anode. They differ in **acid, temperature, power, thickness, and purpose**.

	TYPE I — CHROMIC	TYPE II — SULFURIC	TYPE III — HARDCOAT (THIS MANUAL)
Electrolyte	Chromic acid	Sulfuric acid (~165–200 g/L)	Sulfuric acid (~150–200 g/L, sometimes w/ additives)
Typical thickness	~0.5–7.5 μm	~5–25 μm	~25–100+ μm (commonly ~50 μm / 0.002”)
Temperature	Warm-ish	~20–21 °C (68–72 °F)	Cold — ~-5 to +10 °C (often ~0 °C / 28–32 °F)
Voltage / current	Lower	~12–18 V, ~12–18 ASF	Higher & ramped: ~25–60+ V (up to ~75 V), ~24–40 ASF (~2.5–4 A/dm²)
Color / dyeing	Gray, limited	Excellent dyeing	Natural gray/bronze/dark (alloy-dependent); dye dark only
Hardness / wear	Low–moderate	Hard (ceramic)	Very hard — the wear/abrasion grade
Main use	Aerospace — minimal dim. change & fatigue	Decorative + protective workhorse	Maximum hardness & wear — cylinders, pistons, gears, sliders, tooling

★

REMEMBER
Type I = chromic, thin, aerospace/fatigue. Type II = sulfuric, medium, decorative + protective (the standard). Type III = sulfuric *hardcoat*, thick & hard, run *cold and with more power*, for maximum wear. This manual is Type III — the heavy-duty engineering grade of the same family.

TIP
Don't overstate the family differences. Type III is *not* a different acid from Type II — it's the **same sulfuric chemistry run colder and harder**. That's why a shop can run hardcoat from related chemistry but **never in the same tank at the same time** as Type II — the temperature and power are completely different. Beyond the three “types” there are **specialty anodizes** in this series — **BSAA** (boric-sulfuric, an aerospace chromic replacement), **PAA** (phosphoric, for adhesive bonding), and **two-step electrolytic color** for durable architectural bronzes/blacks. Each gets its own manual.

TECHNICAL STUFF — WHAT “COLDER AND MORE POWER” ACTUALLY BUYS YOU
Three knobs separate Type III from Type II: **(1) temperature** (near 0 °C vs ~20 °C) keeps the acid from re-dissolving the oxide, so it grows dense and hard; **(2) current density** (~24–40 ASF vs ~12–18) grows it faster and thicker; **(3) voltage** climbs much higher (and is *ramped up gradually*) because a thick, dense, insulating oxide needs more push to keep current flowing. Turn all three knobs and ordinary anodize becomes hardcoat. The price is far more heat — hence the heavy refrigeration.

CHAPTER 6

ALUMINUM ALLOYS

ANODIZING ONLY WORKS ON ALUMINUM — AND ON HARDCOAT THE ALLOY CAN MAKE OR BREAK THE PART

Anodizing only works on **aluminum** and a handful of other “valve metals” (titanium, magnesium, tantalum, niobium — each with its own process). On this line, it’s aluminum. But “aluminum” isn’t one thing — it’s dozens of **alloys**, and on hardcoat the alloying elements matter **even more** than on Type II, because the high power and long run amplify every alloy weakness.

WROUGHT VS. CAST

- **Wrought alloys** (rolled/extruded sheet, plate, bar, extrusion — the 1xxx–7xxx series) generally hardcoat **cleanly and predictably**. The everyday hardcoat workhorses — **6061, 6063, and 7075** — all hardcoat well.
- **Cast alloys** (die-cast, sand-cast) usually contain **lots of silicon** and often hardcoat **dark, gray/mottled, lower in hardness, and with a real risk of burning** at the high current densities hardcoat uses. Customers are often surprised that a cast part can’t hit the same hardness or color as a wrought part.

THE BIG TROUBLEMAKERS: COPPER AND SILICON (WORSE ON HARDCOAT)

- **Copper (Cu)**: High-copper alloys — the **2xxx series**, e.g., **2024** (common in aerospace) — are **hard to hardcoat well**. Copper-rich phases dissolve and disrupt the oxide, giving **lower hardness, darker/uneven color, and a high risk of burning** at hardcoat current densities. They often need a **gentler ramp, tighter cooling, lower current density, and a reduced thickness target**.
- **Silicon (Si)**: Silicon particles don’t convert to clear, hard oxide — they leave the coating **gray to dark/charcoal**, lower in hardness, and harder to grow evenly. High-silicon casting alloys are the classic “why is my hardcoat dark and soft?” culprit.

THE FRIENDLY ALLOYS

6061 and 6063 (magnesium-silicon) and **7075** (zinc) hardcoat **very well** and are the everyday favorites for wear parts. **5xxx (magnesium)** generally hardcoats well too. These give the best combination of **hardness, thickness, and a clean (if naturally gray/bronze) color**.

ALLOY FAMILY	MAIN ALLOYING ELEMENT	HARDCOAT BEHAVIOR
1xxx (pure Al)	none (99%+ Al)	Hardcoats but soft base; very light color
2xxx	Copper	Difficult — lower hardness, dark/uneven, burn risk
5xxx	Magnesium	Good
6xxx (6061/6063)	Mg + Si	Excellent — the workhorse
7xxx (7075)	Zinc	Very good — hardcoats well
Cast (high-Si)	Silicon	Dark/mottled, lower hardness, burn risk



REMEMBER

The alloy decides how good the hardcoat can be. The hardest, cleanest, most predictable hardcoat comes from **6061/6063/7075** wrought aluminum. **High-copper (2xxx) and high-silicon (cast) alloys are the two problem families** — lower hardness, darker color, and a real **risk of burning** at hardcoat power levels. Always know your alloy *before* you promise hardness, thickness, or color.



TIP

On a tough alloy (2xxx or high-Si cast), tell your lead **before** you run it. The fixes are real but they cost: a **slower voltage ramp, lower current density, colder bath, and sometimes a reduced thickness target** to avoid burning. A hardcoat that burns on a 2024 part is expensive scrap — anodize a sample of the *actual* alloy and lot first when the spec is tight.



FUN FACT

7075 — a high-strength aerospace alloy — both *hardcoats* well and is *used* for the kind of structural and wear parts that need hardcoat, which is why you’ll see a lot of it on this line. The alloy and the finish were practically made for each other.

CHAPTER 7

HOW THE LINE WORKS

WALKING THE LINE — EVERY TANK HAS A JOB, AND THE ORDER IS THE RECIPE

A Type III line is a sequence of tanks. Here's the whole journey and *why each step has to be where it is*. It's the same backbone as Type II, with the **anodize step colder, longer, and far more powerful**, and the **dye/seal steps optional and spec-driven**.

1. **Inspect / Rack** — Clamp the part tight to a rack. Anodizing is all about **electrical contact** (the part is the anode), and on hardcoat the contacts carry **higher current** — so contacts must be **bigger and tighter** than on Type II. A loose contact = no coating, a burned part, or a dropped part in cold acid.
2. **Clean / Degrease** — Remove oils and shop soil. Oxide won't grow evenly through grease.
3. **Etch (caustic)** — A caustic dip that removes the natural oxide skin and evens the surface. For hardcoat the etch is usually **lighter** than for a decorative satin finish, to **preserve dimension** for the (large) coating growth to come.
4. **Rinse** — Wash off caustic before it contaminates the acid steps.
5. **Desmut / Deox** — An acid dip removes the dark **smut** (alloying elements the etch left behind) so the surface is bright and uniform before anodizing.
6. **Rinse** — Clean before the main tank.
7. **ANODIZE** — The main event. Part = anode, **cold dilute sulfuric acid (~0 °C)**, strong refrigeration and agitation, **DC ramped up to high voltage and current density**. Grow the thick, dense, hard oxide.
8. **Rinse** — Gently rinse acid out of the fresh coating.
9. **Dye (optional, dark only)** — Dark color into the pores, *if specified* (Class 2). Most hardcoat skips this.
10. **Rinse** — Remove surface dye so it doesn't bleed.
11. **Seal (optional, by spec)** — Close the pores for corrosion resistance (Class 2). **Often skipped for maximum-wear (Class 1) parts** — the seal-vs-wear tradeoff.
12. **Rinse / Dry** — Final clean-up and gentle dry.
13. **Inspect** — Thickness, dimensions, hardness/wear, color, and (if sealed) seal quality.



REMEMBER — WHY THE ORDER CAN'T MOVE

You can't dye before you anodize (no pores yet). You can't seal before you dye (sealed pores won't take color). You can't anodize over grease or smut (uneven, defective oxide that burns). And you can't grow a good hardcoat without the **cold bath and ramped power** at step 7. The line *is* the recipe.



HEADS-UP — THE LINE IS A TOUR OF HAZARDS

As you walk the line you pass **hot caustic** (etch), **strong acids** (desmut, anodize), **flammable hydrogen gas** (cathode), **higher-current/higher-voltage DC** (rectifier/contacts — greater arc/shock risk than Type II), a **powerful refrigeration system** (the cold bath), **a lot of exotherm/heat to manage** (burn risk), and **near-boiling water** (seal, if used). Each station chapter calls out its own. Respect each tank for what it is.



TIP

When you're learning, walk the empty line with your trainer and just name each tank's *job* and *main hazard* out loud. If you can do that for all twelve — and explain why the anodize tank is cold and high-power — the rest of the manual is mostly detail.

CHAPTER 8

PPE — YOUR DAILY ARMOR

WHAT YOU WEAR, AND WHY EACH PIECE IS THERE

A hardcoat line works with strong sulfuric acid, hot caustic etch, a near-freezing high-power anodize tank, **higher-voltage DC**, near-boiling seal tanks, chemical mist, and dyes. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't.

HAZARD ON THE LINE	PPE THAT PROTECTS YOU
Sulfuric acid (anodize, desmut) — burns, splash, mist	Sealed chemical splash goggles + face shield , chemical-resistant gloves , apron/suit , chemical-resistant boots
Hot caustic etch — burns worse than you'd expect	Same as above; caustic burns are deep and slow to feel at first
Cold anodize tank / refrigeration (~0 °C acid, chilled surfaces/lines)	Insulated chemical gloves; beware cold acid splash and frosted/slick surfaces
Higher-voltage DC & arc at contacts (more volts/amps than Type II)	Dry hands, dry gloves, no jewelry, follow LOTO; greater shock/arc respect
Near-boiling seal tank (~96–100 °C, if used) — scald/steam	Face shield, heat-resistant gloves, long sleeves; mind the steam
Acid mist / fumes	Local exhaust at tanks; respiratory protection per your program
Dyes (staining, some sensitizers)	Gloves; avoid skin/eye contact; check the dye SDS
Slips on wet/cold floors	Non-slip chemical-resistant boots

TIP

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. Rinse your gloves *before* you take them off.

★ REMEMBER

On a hardcoat line, your **eyes and skin** are the most exposed, and you've now got an **electrical** exposure bigger than Type II (more volts and amps) plus a **cold** exposure (near-freezing acid). **Sealed splash goggles and a face shield are not optional** at the working tanks, and **dry hands/gloves around the rectifier and contacts are a hard rule**.

⚠ HEADS-UP

Goggles for chemicals, not just safety glasses. Safety glasses stop flying chips; they do **not** stop an acid or caustic splash from getting into your eyes from the side. At every chemical tank, sealed splash goggles (and a face shield when charging, dumping, racking, or handling parts) are required.

⚠ HEADS-UP

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting chemicals — especially dyes and (if used) nickel-bearing seal salts. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 9

EMERGENCY & FIRST-RESPONSE

KNOW THESE COLD BEFORE YOU TOUCH A TANK

Know these *before* you need them. In an emergency, seconds matter and you won't have time to read.

SITUATION	WHAT TO DO — IMMEDIATELY
Acid or caustic on skin	Flush immediately with water for at least 15 minutes . Remove contaminated clothing while flushing. Get medical attention — caustic burns especially can be deep even when they don't hurt at first.
Acid or caustic in eyes	Go straight to the eyewash , hold eyes open, flush at least 15 minutes , and get medical help. Don't stop early.
Large splash / drench	Use the safety shower — full 15 minutes, clothing off.
Cold-acid splash (near-freezing bath)	Treat as a chemical burn first — flush 15 min (the cold does not make the acid less corrosive); then address any cold-related skin injury.
Hot seal-tank scald (if sealing)	Cool the burn with water; do not pop blisters; get medical attention for anything beyond minor.
Acid spill	Don't play hero — alert others, contain only if trained, use the spill kit, follow your shop's spill plan. Never mix acids and caustics during cleanup.
Electrical / arc at rectifier or contacts	Hardcoat runs higher voltage/current — arc and shock are more dangerous here. Don't touch — de-energize via the rectifier controls / LOTO. Never reach into a tank with the current on.
Hydrogen gas	Keep ignition sources (flames, sparks, grinding) away from the cathode/tank area; ensure ventilation is running.
Fluoride exposure (fluoride desmut / cold nickel-fluoride seal)	Fluoride burns are deep, delayed, and systemically toxic — flush, then apply the specific first-aid in the SDS (often calcium gluconate gel) and get medical help immediately. Ordinary flushing alone may not be enough.

⚠

HEADS-UP — THE GOLDEN RULE FOR ACID MAKE-UP

Always add **ACID to WATER**, never **water to acid**. (“Do as you *oughta*: add acid to water.”) Adding water to concentrated sulfuric acid can flash-boil and erupt the acid out of the container. Bath make-up/additions are usually authorized-personnel work — but every operator must know this rule.

⚠

HEADS-UP — LOCKOUT/TAGOUT (LOTO)

LOTO is mandatory for *all* rectifier and electrical maintenance — and on hardcoat the rectifier is **bigger and runs at higher voltage**, so the stakes are higher. Physically lock the power off and tag it so no one can switch it back on while someone is working on it. “I’ll just be a second” is how people get electrocuted.

★

REMEMBER

Know where the **nearest eyewash, safety shower, and spill kit** are at *every* station before you start your shift. In an acid/caustic emergency you won't have time to look. **Eyewash and shower stations must always be unblocked.**

CHAPTER 10

THE SAFETY PHILOSOPHY OF A HARDCOAT LINE

THE MINDSET THAT TIES EVERY STATION TOGETHER

1. TWO CORROSIVES, MIRROR-IMAGE.

The **etch** is **caustic** (high pH); the **desmut** and **anodize** are **acid** (low pH). Both burn. Treat high pH and low pH with equal respect — and never let them mix outside a controlled neutralization.

2. THE PART IS LIVE — AND MORE LIVE THAN TYPE II.

Because the part is the anode, there is **DC current at the work** and at the contacts. Hardcoat runs **higher voltage and current**, so the shock and arc hazard is bigger. Wet hands + current + acid is a bad combination.

3. COLD IS A HAZARD TOO.

The anodize tank and its refrigeration run **near freezing**. Cold acid is **just as corrosive** as warm acid; chilled surfaces and lines can be slick or cause cold injury. And paradoxically, the cold tank makes *enormous heat* internally — see #4.

4. HEAT AND “BURNING” ARE CONSTANT THREATS.

Hardcoat dumps a lot of electrical energy into the part. If cooling falters or current is pushed too fast, the part **burns** (locally overheats — powdery, charred, scrapped) and the bath can overheat. Watch the cooling the *entire* run.

5. HYDROGEN IS FLAMMABLE.

The cathode bubbles off hydrogen gas. Ventilation on, ignition sources away.

6. ENGINEERING CONTROLS FIRST, PPE SECOND.

Tank exhaust, refrigeration, agitation, lids, and splash guards do the heavy lifting; PPE is your last layer, not your first.

7. WHEN IN DOUBT, STOP AND ASK.

The unsafe operator isn't the one who asks questions — it's the one who guesses.

• THE HAZARD FAMILIES TO INTERNALIZE

1. **Corrosives (acid AND caustic)**. Sulfuric (anodize/desmut) and caustic (etch) burn skin and eyes — cold acid included. *Defense*: goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
2. **Thermal — both extremes**. Near-freezing anodize bath/refrigeration *and* (if sealing) near-boiling seal, plus the **internal exotherm/burn risk** of the high-power process. *Defense*: insulated gloves, steam awareness, and *watch the cooling*.
3. **Electrical & gas. Higher-voltage, higher-current DC** (bigger arc/shock risk than Type II); flammable hydrogen at the cathode. *Defense*: LOTO, dry hands, no ignition sources, ventilation.

★ REMEMBER

Corrosives first, thermal (cold *and* hot *and* burn-risk) second, electrical/gas third. If you can name the hazard family at the tank in front of you, you already know most of what you need to protect yourself.

☀ FUN FACT

Many veteran hardcoat operators can *hear and smell* a bath going wrong — a change in the bubble sound, a whiff of burning — before any instrument confirms it. But notice the lesson hidden there: *they got to be veterans by never getting hurt*. Skill and safety grow together.

★ REMEMBER

You now have the whole foundation: what anodizing is (and how it's *not* plating), the part-is-the-anode/grow-oxide idea, the **cold-makes-hard** principle, the **seal-vs-wear tradeoff**, **big dimensional growth**, the Type family, alloy effects, how the line works, what to wear, and the safety mindset. Every station chapter builds on this. Turn the page knowing the *why*, and the *how* will make a lot more sense.

PART TWO — STATION BY STATION
STATION 01 OF 12

INSPECTION & RACKING

“IN ANODIZING, THE PART IS THE WIRE. ON HARDCOAT IT CARRIES A LOT OF CURRENT — A LOOSE PART IS A DEAD (OR BURNED) PART.”

PURPOSE & THE “WHY”

Inspect the incoming aluminum part, confirm the **alloy and the spec** (Type III, thickness, **class** — MIL-A-8625 Class 1 unsealed or Class 2 sealed/dyed, AMS callouts), and **rack** it so it makes firm, **high-current** contact and hangs correctly. Because the part **is the anode**, every bit of current flows *through the rack contact into the part* — and hardcoat current is **higher** than Type II, so a contact that was fine for decorative anodize can **overheat and burn** on hardcoat. A loose or undersized contact means **no coating, a thin coating, a burned contact spot, or a dropped part in cold acid**. Racking also decides *where* the un-anodized contact mark lands, how gas escapes, and how solution drains.



REMEMBER — THE CONTACT MARK IS UNAVOIDABLE, AND BIGGER HERE

Wherever the rack grips the part, **no oxide grows** (that spot carries current, so it can't insulate itself). On hardcoat the contacts are **larger and more numerous** to carry the higher current without burning — so the contact marks are bigger too. Place them where they won't matter and make them **very tight**.

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / print	Confirm alloy , Type III, thickness, class (1 unsealed / 2 sealed-dyed)
Rack material	Titanium or aluminum	Ti racks don't anodize away; Al racks anodize and need stripping
Contact size/number	Larger & more contacts than Type II	Higher hardcoat current → undersized contact burns
Contact pressure	Firm, spring-loaded / bolted	Loose = burning, no coat, dropped parts
Contact location	Hidden/non-critical area	The contact point won't anodize (bare mark)
Masking	Hardcoat-rated plugs/tape/lacquer	Threads, grounds, bearing fits, press-fits (growth is large)
Orientation	Allow gas escape & drainage	Avoid trapped air pockets (un-anodized voids)



HEADS-UP

Racks and parts have **sharp edges and pinch points**, and you'll be handling them near chemical tanks and a **higher-current bus**. Watch your hands; keep hands dry near the contacts. A part that drops because of a weak rack becomes a **splash hazard** in cold acid, not just a scrap part.



TIP — SIZE THE CONTACT FOR THE CURRENT

On hardcoat, the rule is **more and bigger contacts than you'd use for Type II**. Each contact has to carry its share of a much larger total current without local overheating. Undersized contacts are the #1 cause of **contact-area burning** on hardcoat. Titanium racks hold contact load best and don't need stripping; aluminum racks hardcoat right along with the part (insulating the contact) and must be **stripped/re-pointed** often.

STEP-BY-STEP

1. Read the traveler (alloy, Type III, thickness, **class**, masked areas).
2. Inspect for scratches/gouges/prior anodize — hardcoat *reveals* flaws, doesn't hide them.
3. Plan **hidden, generously sized contact points** that can carry full hardcoat current.
4. Mask no-anodize areas per print (remember the large growth).
5. Rack firmly — good metal-to-metal contact, correct orientation.
6. Verify no parts touch each other (shadowing).
7. Handle by the rack from here on.

SIGN-OFF — STATION 01

Teach-Back: Why contact matters even more on hardcoat (higher current → burn risk); why contacts are bigger/more numerous; why the contact point won't anodize; why titanium racks are preferred; why orientation (gas/drainage) matters; two no-anodize areas that get masked and why.

"I verified the alloy and class, made firm, generously sized contacts in a non-critical area, masked all no-anodize features for the large hardcoat growth, racked for gas escape and drainage with no parts touching, and handled parts by the rack."

Trainee: _____ Trainer: _____ Date: _____

STATION 02 OF 12

CLEAN / DEGREASE

“ANODIZE OVER OIL AND YOU’VE GROWN A DEFECT, NOT A COATING — AND ON HARDCOAT THAT’S AN EXPENSIVE DEFECT.”

PURPOSE & THE “WHY”

Get the part **chemically clean** — no oil, grease, fingerprints, or shop soil — so the oxide grows evenly. Oxide can only grow uniformly from clean aluminum; oil and soil cause **uneven growth, streaks, soft spots, and burning** under hardcoat’s high power. Cleaning is usually a **mild alkaline (non-etching) soak cleaner** — formulated *not* to attack the aluminum (we save the controlled metal removal for the etch). The free, foolproof check is the **water-break test**.



TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous, unbroken film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds; use it on every part.

FAIL - water beads up

```
+-----+
| o o o | X
| o o |
+-----+
```

Oil remains -> RE-CLEAN

PASS - water sheets evenly

```
+-----+
|#####| OK
|#####|
+-----+
```

Surface clean -> proceed

PARAMETER	RANGE	WHAT TO WATCH
Type	Mild non-etching alkaline cleaner	Must not attack aluminum
Temperature	~50–70 °C (120–160 °F)	Per cleaner TDS
Time	~3–10 min	Heavy oil longer
Concentration	Per TDS	Titrate regularly
pH	Mildly alkaline	Still a skin irritant/burn risk
Water-break test	Pass = continuous sheet	The only reliable field check



HEADS-UP — HOT ALKALINE SOLUTION

Even a “mild” cleaner is hot and alkaline: splashes irritate/burn skin and eyes, and steam irritates airways. Splash goggles, gloves, apron. Skin → flush 15 min; eyes → eyewash 15 min and get help.

STEP-BY-STEP

1. Check tank temperature and concentration are in range. 2. Immerse the racked load fully; start the timer; confirm agitation. 3. Withdraw slowly, draining back into the tank. 4. Run the **water-break test** — continuous sheet → proceed; beading → re-clean. 5. Move promptly to the etch (don’t let a clean part sit).

SIGN-OFF — STATION 02

Teach-Back: What the cleaner removes (oil/soil) vs. what it must *not* do (attack aluminum — that’s the etch); correctly perform/read a water-break test; the caustic/irritant hazard and PPE; why a non-etching cleaner is used here.

“The part passed the water-break test, the cleaner was in range and at temperature, and I wore required PPE.”

Trainee: _____ Trainer: _____ Date: _____

STATION 03 OF 12

ETCH (CAUSTIC, LIGHT FOR HARDCOAT)

“FRESHEN THE SURFACE — BUT ON HARDCOAT WE ETCH LIGHTLY TO SAVE DIMENSION FOR THE BIG COATING TO COME.”

PURPOSE & THE “WHY”

Remove the natural oxide skin and even out the surface in a **caustic (sodium hydroxide) etch**, giving a fresh, uniform surface for anodizing. Aluminum always carries a thin, uneven natural oxide and fine surface defects; the caustic etch dissolves a controlled amount of aluminum to clean that up. **The hardcoat difference:** because the coating is *thick* and grows the part *significantly* (Ch 4), shops usually run a **lighter etch for hardcoat** — just enough to clean and uniform the surface — to **preserve dimensional tolerance** for the large growth ahead. Etch too much and you’ve lost dimension *before* the coating even adds its own.

★

REMEMBER — ETCH LIGHT TO PROTECT TOLERANCE
On Type II the etch sets a decorative satin texture. On hardcoat, appearance is rarely the point and **dimension is precious**, so the etch is typically **lighter and more controlled** — just enough to clean and uniform the surface. Heavy etching wastes metal you need for tolerance.

PARAMETER	RANGE	WHAT IT DOES / WATCH
Chemistry	Sodium hydroxide, often with additives	Dissolves oxide + a little aluminum
Concentration	~40–60 g/L NaOH	Higher = faster/more aggressive
Temperature	~50–65 °C (120–150 °F)	Hot — burns and speeds etch
Time	Short (often ~0.5–2 min for hardcoat)	Keep light to preserve dimension
Result	Clean, uniform surface + smut	Smut is normal — removed at desmut

⚠

HEADS-UP — HOT CAUSTIC IS ONE OF THE WORST BURNS ON THE LINE
Caustic burns are **deep, get worse with time, and may not hurt much at first** — which makes people under-react. Full PPE: sealed goggles, face shield, elbow-length chemical gloves, apron, boots. Skin → flush **15+ minutes**; eyes → eyewash 15 min and get medical help immediately. Never assume “it’s just a little.”

⚠

HEADS-UP — CAUSTIC + ALUMINUM MAKES HYDROGEN
Caustic attacking aluminum releases **hydrogen gas** (flammable). Keep ignition sources away and ventilation running.

STEP-BY-STEP

1. Confirm concentration and temperature in range. 2. Immerse; start the timer for the **specified (usually short) etch time** — controls cleanliness and metal loss. 3. Agitate gently / per procedure. 4. Withdraw and drain back into the tank — the part will look **dark/smitty** (normal). 5. Move **directly to rinse** — don’t let caustic dry on the part.

SIGN-OFF — STATION 03

Teach-Back: What the etch removes (oxide + a little aluminum); why hardcoat uses a *lighter* etch (preserve dimension for large growth); why the part comes out dark/smitty and what fixes it (desmut); the caustic-burn hazard and first response (15-min flush, get help).

“I etched lightly for the specified time at the correct concentration/temperature, protected dimension for the hardcoat growth, moved straight to rinse, and wore full PPE for hot caustic.”

Trainee: _____ Trainer: _____ Date: _____

STATION 04 OF 12

RINSE (THE FIREWALL)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

PURPOSE & THE “WHY”

Wash the **caustic film** off the part before it reaches the acidic desmut and anodize chemistry. The etch is caustic (high pH); desmut and anodize are acidic (low pH). Carry caustic drag-out forward and it **neutralizes and contaminates the acid baths**, wastes chemistry, and causes staining — and on a hardcoat bath, contamination that disturbs the carefully held cold chemistry is especially costly. The rinse knocks drag-out down by dilution. Cleanliness is tracked by **conductivity** (µS/cm) — *lower = cleaner*.



TIP — COUNTER-FLOW RINSING

The professional approach is a **counter-current** cascade: parts move forward, water flows backward, so the cleanest water touches the part last. It saves enormous water — important on a line with this many rinses.

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating needed
Time	30–60 s with agitation	Longer for cavities/blind holes
Water source	Municipal or DI	DI preferred near anodize/seal
Conductivity	< 500 µS/cm (target < 300)	Monitor at shift start
Flow	Continuous overflow / counter-current	Prevents stagnation



HEADS-UP

This rinse carries dragged-in caustic and starts mildly alkaline. Goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses.**

DO

- Check conductivity at shift start; tell your lead if near/above limit.
- Transfer promptly from the etch — don’t let caustic dry.
- Immerse fully with agitation 30–60 s (longer for cavities).
- Lift and drain over the rinse tank before moving on.

TOP MISTAKES

- Treating the rinse as a quick dunk.
- Ignoring conductivity.
- Letting parts drip-dry between etch and rinse.
- Not flushing cavities; skipping the drain step.

SIGN-OFF — STATION 04

Teach-Back: Why carrying caustic into the acid stages causes problems; define drag-out / drag-in; what conductivity tells you and which way is “clean”; the conductivity limit and target.

“Rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn.”

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm

STATION 05 OF 12

DESMUT / DEOX

“ETCHING UNCOVERS THE ALLOY’S LEFTOVERS. DESMUT WASHES THEM AWAY — AND ON TOUGH HARDCOAT ALLOYS THERE’S MORE OF THEM.”

PURPOSE & THE “WHY”

Remove the dark **smut** the etch left behind — the alloying elements (copper, silicon, iron, manganese) that didn’t dissolve in caustic — using an **acid desmut/deox**, leaving a bright, uniform surface for anodizing. When caustic dissolves aluminum, the **other elements in the alloy stay behind as a dark film (smut)** on the surface. Anodize over smut and you get **dark streaks, dull patches, soft spots, and — under hardcoat’s high power — burning**. The desmut chemistry depends on the alloy: simple alloys may need only **nitric** or **sulfuric**; **high-silicon (cast) and high-copper (2xxx) hardcoat alloys** — which leave heavier smut — often need a **fluoride-bearing (or proprietary) deox** to cut silicon/copper smut.

★

REMEMBER

Smut after etch is normal, and the tough hardcoat alloys (2xxx, high-Si cast) leave more of it. Desmut’s whole job is to remove it. A part that goes into the cold hardcoat tank still gray/smitty will come out streaky, dull, soft — and is a prime candidate to **burn**. Desmut is the bright-surface gatekeeper.

PARAMETER	RANGE / PRACTICE	NOTES
Chemistry	Nitric and/or sulfuric; fluoride-bearing/proprietary deox for high-Si/Cu	Match to the alloy
Concentration	Per product TDS	Titrate regularly
Temperature	Often ambient-warm	Per TDS
Time	~0.5–3 min	Until smut is gone, no longer
Result	Bright, uniform, active surface	No gray film remaining

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HEADS-UP — STRONG ACID (AND POSSIBLY FLUORIDE)

Desmut is acidic; **fluoride-bearing desmuts are especially hazardous** (hydrofluoric-type chemistry can cause deep, delayed burns and is systemically toxic). Know exactly which chemistry your tank uses, read its SDS, wear full acid PPE, and know the **specific first-aid** (fluoride burns may require **calcium gluconate** per your SDS/medical plan). **Add acid to water** for any make-up.

STEP-BY-STEP

1. Confirm which desmut chemistry the tank is (and its specific hazards/first-aid). 2. Check concentration and temperature in range. 3. Immerse for the specified short time — just until smut clears. 4. Withdraw, drain, move to rinse. 5. Don’t over-dip — excessive desmut can etch/pit some alloys and waste dimension.

SIGN-OFF — STATION 05

Teach-Back: What smut is and where it came from (alloying elements left by the etch); why anodizing over smut causes defects/burning on hardcoat; that desmut chemistry depends on the alloy (fluoride for silicon/copper); the special hazard/first-aid of a fluoride-bearing desmut.

“I removed the smut with the correct desmut for the alloy, did not over-dip, and understand the specific hazard and first-aid for this tank (including fluoride if applicable).”

Trainee: _____ Trainer: _____ Chemistry: _____

STATION 06 OF 12

RINSE (PRE-ANODIZE)

“THE LAST CLEAN HANDOFF BEFORE THE MAIN EVENT.”

PURPOSE & THE “WHY”

Remove desmut acid before the part enters the cold hardcoat tank. Carrying desmut drag-out into the anodize tank introduces unwanted ions (and, with a fluoride desmut, **fluoride contamination** that can roughen/pit the hardcoat). A clean rinse protects the carefully controlled cold anodize bath.

PARAMETER	RANGE	NOTES
Temperature	Ambient	—
Time	30-60 s, agitation	Longer for cavities
Water	DI preferred	Cleaner handoff to anodize
Conductivity	Per shop spec	Lower = cleaner
Flow	Counter-current overflow	—



HEADS-UP

Same rinse rules: drag-in acid, wet floors, goggles/gloves. If the prior tank was a **fluoride** desmut, this rinse water carries fluoride — handle and dispose accordingly.



TIP

Move from this rinse into the cold anodize tank **promptly and consistently**. A long, variable dwell in air or rinse before anodizing can cause part-to-part thickness/hardness differences, and a warm or dried part entering the cold bath disturbs temperature control. Consistency is quality.

STEP-BY-STEP

- Confirm flow and conductivity.
- Rinse the full time with agitation.
- Drain over the tank.
- Move straight to anodize.

TOP MISTAKES

- Quick dunk instead of a real rinse.
- Inconsistent dwell before anodize.
- Ignoring conductivity.
- Forgetting fluoride drag-in handling.

SIGN-OFF — STATION 06

Teach-Back: Why a clean handoff to the cold hardcoat tank matters; what fluoride drag-in could do.

“The part was fully rinsed, conductivity was acceptable, and I moved promptly and consistently to the anodize tank.”

Trainee: _____ Trainer: _____ Date: _____

STATION 07 OF 12 • THE MAIN EVENT

ANODIZE — THE MAIN TANK

WHERE THE ALUMINUM'S OWN SURFACE BECOMES A THICK, DENSE, VERY HARD CERAMIC — AND WHERE THE PART IS THE ANODE

Take a breath. Up to now, every station has had one job: get the part clean, even, and ready. **Station 07 is where the hardcoat is actually created** — the tank that defines this whole line.

The big idea in one sentence: **we run the aluminum part as the anode in near-freezing, dilute sulfuric acid and ramp DC current up to high voltage and high current density to grow a thick, dense, very hard aluminum-oxide layer out of the part's own surface.** The cast:

- **Anode** — *your aluminum part (positive)*. Where the oxide grows.
- **Cathode** — **lead or aluminum (6063) plates** (negative). They carry current and bubble off hydrogen; they do **not** feed metal into the part. (*Hardcoat often uses more cathode area than Type II to handle the higher current.*)
- **Electrolyte** — **dilute sulfuric acid (~150–200 g/L, ~15–20% v/v)**, held **very cold (~–5 to +10 °C; often ~0 °C / 28–32 °F)**, with **strong agitation and powerful refrigerated cooling**.



REMEMBER — THE HARDCOAT TANK IS THE MIRROR IMAGE OF A PLATING TANK

In plating, the part is the **cathode** and metal lands on it. Here the part is the **anode** and **oxide grows out of it**. The counter-electrodes (cathodes) don't supply the coating — all the coating comes from the part's own aluminum plus oxygen from the bath. Forget this and the whole tank behaves like a mystery.



REMEMBER — COLD + POWER IS THE RECIPE

A hardcoat tank is “a Type II tank turned **much colder** and given **a lot more power**.” Cold makes the oxide **dense and hard**; high voltage and current density make it **thick**. Lose the cold, and you don't get hardcoat — you get a soft, burned mess.



HEADS-UP — SULFURIC ACID + HIGH-POWER DC + HYDROGEN + HEAVY REFRIGERATION, ALL AT ONCE

This tank holds **strong sulfuric acid** (burns, splash, mist — *cold acid burns just as badly*), runs **higher-voltage, higher-current DC** than Type II (greater shock/arc hazard at contacts and bus), bubbles **flammable hydrogen** off the cathodes, **self-heats hard** (the refrigeration *must* keep up or the coating burns and the bath overheats), and relies on a **powerful refrigeration system** (chilled lines/surfaces).

Engineering controls — **exhaust, refrigeration, agitation, splash control** — must be working before you run it. Full acid PPE. No ignition sources.

• WHAT MAKES HARDCOAT PARTICULAR

The same sulfuric chemistry as Type II demands much more care here: it must be **kept near 0 °C** (warm bath = soft/powdery/“burned” coating — you lose the hardcoat), the **voltage is ramped up** to hold a steady high current as the thick, insulating oxide builds, it's sensitive to **chloride contamination** (pitting/burning), it **self-heats hard** so cooling and agitation are make-or-break, and it grows the coating **half into the part** — a *large* dimensional change on hardcoat. Get the cold, the ramp, and the cooling right and you get a flawless engineering coating; miss any of them and you get scrap.

• THE BATH MAKE-UP — KNOW THE RANGES

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES / WATCH
Sulfuric acid (H ₂ SO ₄)	~150–200 g/L (~15–20% v/v)	The electrolyte; grows + (only slightly, when cold) dissolves the oxide
Temperature	~-5 to +10 °C; commonly ~0 °C (28–32 °F)	The critical control — cold = dense/hard; too warm = soft/burned
Voltage (DC), ramped	~25–60+ V (up to ~75 V on thick coats)	Ramped up gradually ; rises as the thick insulating oxide builds
Current density	~24–40 ASF (~2.5–4 A/dm ²)	Higher than Type II — sets growth rate; must not exceed burn limit
Coating thickness	~25–100+ µm (0.001–0.004"+); commonly ~50 µm (0.002")	Per spec; controlled by current × time
Time	~30–90+ min (longer than Type II)	Thick coats run long; cooling must hold the whole time
Cathodes	Lead or aluminum 6063 (often more area)	Carry current; bubble H ₂ ; do not feed metal
Agitation	Strong air/solution agitation	Even temp & fresh acid at the surface — critical to avoid burning
Cooling	Powerful refrigeration / heat exchanger	Bath self-heats hard — must be cooled aggressively
Dissolved aluminum (Al ³⁺)	Low, per spec (e.g., ≤ ~12–15 g/L)	Too much dulls/softens & slows the bath
Chloride contamination	Very low (≤ ~50 ppm typ.)	Chloride causes pitting/burning — watch it



REMEMBER — THE NUMBERS TO CARRY

Sulfuric ~150–200 g/L, temp ~0 °C (28–32 °F), ~25–60+ V (ramped), ~24–40 ASF, ~25–100+ µm (commonly 50 µm / 0.002"), ~30–90+ min, lead/6063 cathodes, strong agitation + powerful cooling. Compared to Type II: *colder, more volts, more amps, thicker, longer.*



TECHNICAL STUFF — RAMPING THE VOLTAGE (WHY YOU CAN'T JUST SLAM IT ON)

A thick, dense hardcoat is **insulating**, so as it grows, the rectifier must push **more and more voltage** to keep the same current flowing through it. You **ramp the voltage up gradually** at the start (and let it climb during the run) so current density stays in the target band without spiking. Slamming full power onto a bare part **crowds current at edges and contacts and burns them**. A controlled ramp is the difference between a flawless hardcoat and a charred one.



TECHNICAL STUFF — WHY TEMPERATURE IS EVERYTHING ON HARDCOAT

Warm bath → the acid dissolves the growing oxide too fast → a **soft, powdery, thin, or "burned"** coating with poor hardness and wear. Cold bath → the acid hardly touches the oxide → a **hard, dense, thick** coating. Type II tolerates ~20 °C; **hardcoat must be held near 0 °C**. If the refrigeration falls behind even a few degrees mid-run, the coating quietly stops being hardcoat. **Watch the temperature the entire run.**



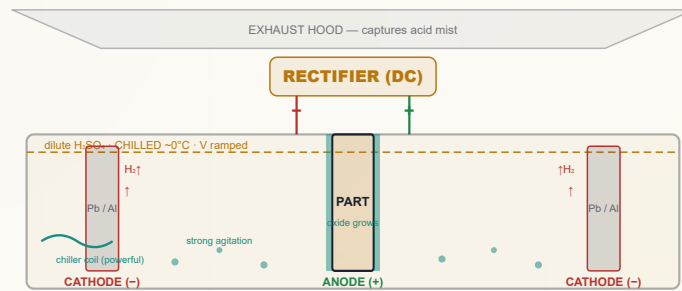
TECHNICAL STUFF — BURNING, THE HARDCOAT FAILURE MODE

"Burning" is local runaway: a spot (often an edge, corner, or contact) gets too much current density and too little cooling, the oxide there overheats, and it turns **powdery, dark, and scrapped** — and can even melt or pit the part. Hardcoat's high power makes burning *the* defect to fear. Defenses: **strong agitation, aggressive cooling, a controlled voltage ramp, generous contacts, and not exceeding the current-density limit** (lower it for tough 2xxx/cast alloys).



TECHNICAL STUFF — DISSOLVED ALUMINUM AND CHLORIDE

A *little* dissolved aluminum is normal, but **too much** softens the coating, slows growth, and causes streaks — the bath eventually needs partial decant/replacement. **Chloride** (from tap water, hand sweat, or drag-in) is the silent killer: even tens of ppm can cause **pitting and burning**. Use DI water and keep contamination out.



Key params: H2SO4 150-200 g/L • ~0 °C (28-32 °F) • 25-60+ V ramped • 24-40 ASF • 25-100+ um • strong agit + heavy cooling. Note the **reversed polarity** vs. a plating cell: the part is the ANODE (+).

• STANDARD OPERATING PROCEDURE

1. **Confirm the part is clean, lightly etched, desmutted, rinsed, and racked tight with generous contacts** — straight from the rinse, no long air dwell.
2. **Verify engineering controls FIRST:** refrigeration running and bath **at temperature (~0 °C)**, strong agitation on, exhaust on, splash control in place, full acid PPE on.
3. **Check bath readiness:** sulfuric in range (per lab), **temperature in range (the make-or-break check)**, dissolved-aluminum and chloride within limits, cathodes present/sound and sufficient.
4. **Confirm rectifier polarity: PART = ANODE (POSITIVE).** (*This is the opposite of plating — verify it every time.*)
5. **Set/plan current for the part's surface area** to hit the target current density (~24-40 ASF) — and the **voltage ramp** schedule.
6. **Calculate the time** for target thickness (current density × time) and set the timer/amp-hour counter. Thick hardcoat runs long.
7. **Energize and RAMP the voltage up gradually** per procedure; watch that **voltage rises** to hold current as the thick insulating oxide grows (a large voltage rise is normal on hardcoat).
8. **Anodize for the calculated time**, monitoring **temperature (constantly)**, **current/voltage, agitation, and cooling** the *whole* run — hardcoat is *not* set-and-forget.
9. **At target, stop current, withdraw, drain over the tank**, and move to the post-anodize rinse — ready for (optional) dye/seal, or for Class 1 straight toward inspection.
10. **Log everything:** time, current/voltage (and ramp), temperature, thickness checks, additions.



HEADS-UP — VERIFY POLARITY EVERY SINGLE TIME

Because everyone trained on plating expects the part to be the **cathode**, the most dangerous habit on this line is assuming polarity. **The part is the ANODE (+).** Reversed polarity gives no coating (and can damage parts/cathodes) — and on hardcoat's high power, a wiring error is more dangerous. Check it before you energize.



HEADS-UP — WATCH THE COOLING AND THE RAMP THE WHOLE LONG RUN

A hardcoat run is **30-90+ minutes** and the bath self-heats hard the entire time. If cooling falls behind or you ramp too fast, the coating **burns** and the bath degrades. **Do not set-and-forget a hardcoat run.**

COMMON DEFECTS — WHAT YOU’LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Burned (powdery, dark, charred, esp. edges/contacts)	Cooling lost; CD too high; ramped too fast; poor agitation; undersized contacts	Restore cooling/agitation; lower CD; slow ramp; add/bigger contacts; verify temp
Soft / low-hardness coating	Bath too warm; high dissolved Al; over-sealed	Restore cold temp; lab-check Al; reconsider seal (Station 10)
Thin / no coating	Bad contact; wrong polarity; low current; low acid; ramp stalled	Re-rack tight; verify part = anode ; check current/acid/ramp
Pitting	Chloride contamination; bath impurities	Lab-check chloride; use DI; treat bath per chemist
Dull, dark, gray, mottled	High-Si/Cu alloy; high dissolved Al; smut not removed	Confirm alloy; lab-check Al; verify desmut
Thickness/hardness varies part-to-part	Mixed alloys; inconsistent temp/CD/dwell	Rack like-with-like; standardize ramp/timing; verify temp
Voltage spikes / won’t hold current	Poor contact; failing/insufficient cathodes; low conductivity	Check contacts/cathodes; lab-check bath
Cracking / crazing of thick coat	Very thick coat + thermal stress (can be inherent)	Confirm thickness vs spec; discuss with chemist

INTEGRATED SAFETY — THIS STATION DEMANDS TOTAL RESPECT

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HEADS-UP — SULFURIC ACID (THE HEADLINE)
Corrosive, splash and **acid mist** hazard — and **cold acid is just as corrosive**. Engineering controls (exhaust, splash guards) first, PPE always (sealed goggles, face shield, gloves, apron, boots). Any skin/eye contact → flush **15 minutes**, get medical attention. **Acid to water** for make-up.

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HEADS-UP — HIGHER-VOLTAGE DC & FLAMMABLE HYDROGEN
Hardcoat runs **more volts and amps** than Type II — bigger shock/arc hazard at contacts and bus bars. **LOTO** for all electrical work, dry hands only. Hydrogen off the cathodes is **flammable** — ventilation on; no flames, sparks, or grinding nearby.

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HEADS-UP — REFRIGERATION, EXOTHERM, AND BURNING
The bath relies on a **powerful refrigeration system** (chilled surfaces/lines — cold-contact and slip risk) and **self-heats hard** the whole run. If cooling fails or temperature climbs out of range mid-run, the coating **burns** and the bath degrades. **Watch the cooling the entire run.**

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part, and which charge? (*Anode, +.*)
- Where does the coating come from? (*The part’s own Al + O₂ — not the cathodes.*)
- The headline numbers (acid, temp, V, CD, thickness, time)?
- Why must the bath be *near freezing*, and what if it warms?
- Why do you *ramp* the voltage?
- What is “burning” and how do you prevent it?
- The safety families? (*Acid, high-power DC, hydrogen — plus cold/exotherm.*)

TOP MISTAKES TO AVOID

- Assuming the part is the cathode (it’s the **anode**).
- Running with failing/insufficient cooling.
- Ramping the voltage too fast / slamming on full power.
- Undersized or loose contacts (burning).
- Ignoring chloride / dissolved-Al limits.
- Mixing alloys on a rack and expecting matched hardness.
- Long, variable dwell before anodizing.

SIGN-OFF — STATION 07

“I can independently: verify refrigeration, agitation, exhaust, and PPE before running; verify bath readiness (acid, **temperature near 0 °C**, dissolved Al, chloride per lab); confirm the part is racked as the **ANODE** (positive) with generous contacts; set the correct current density and **ramp the voltage gradually**; control thickness by current × time; recognize **burning**, soft-coat, pitting, and thin-coat defects; and state the acid, high-power electrical, hydrogen, and cold/exotherm hazards. I understand chemistry adjustments are made only by authorized personnel.”

Operator: _____ Date: _____ Trainer: _____ Date: _____

STATION 08 OF 12

RINSE (POST-ANODIZE)

“GENTLE NOW — THE FRESH COATING IS ABSORBENT AND COLD.”

PURPOSE & THE “WHY”

Rinse the sulfuric acid out of the freshly anodized part **without contaminating the fresh coating**, before any dye or seal (or, for Class 1, before final handling). The fresh anodic oxide is **porous and absorbent** — it will soak up whatever it touches. Leftover acid interferes with any dyeing and sealing. The rinse must be **clean (DI)** and — *if the part will be dyed* — **not too warm**, because warm water can begin to seal the pores early and **block dye uptake**. (For Class 1 unsealed/natural parts, simply rinse the acid out cleanly.)



REMEMBER — DON'T PRE-SEAL BY ACCIDENT (IF DYEING)

If the part is to be dyed, **use cool/ambient DI rinse here**. Hot rinse water can start closing the pores, and a partially sealed part **won't dye properly**. Save the heat for the actual seal step. (For natural/unsealed Class 1 parts there's no dye to protect, but a clean DI rinse is still required.)

PARAMETER	RANGE	NOTES
Water	DI strongly preferred	Pores absorb impurities
Temperature	Cool / ambient (before dye)	Avoid premature sealing
Time	1-3 min, gentle agitation	Flush acid from pores
Conductivity	Low (per spec)	Confirms acid removed



HEADS-UP

This rinse carries dragged-out **sulfuric acid** — mildly acidic, slip hazard, goggles/gloves. Don't let acid-laden rinse go untreated to drain (Part Three / waste).

STEP-BY-STEP

- Confirm DI flow, cool temperature.
- Rinse gently, full time.
- Drain.
- To dye (Class 2/dyed); to seal (if sealed); or toward inspection (Class 1 natural).

TOP MISTAKES

- Using hot water before dye (pre-seals pores).
- Using hard tap water (impurities in the coating).
- Short rinse leaving acid behind.

SIGN-OFF — STATION 08

Teach-Back: Why the fresh coating is absorbent; why cool DI before dye; what hot water would do prematurely; what to do for a Class 1 natural part.

“I rinsed with cool DI to flush acid from the fresh coating without pre-sealing, and moved appropriately to dye, seal, or inspection per the part's class.”

Trainee: _____ Trainer: _____ Date: _____

STATION 09 OF 12 • OPTIONAL

DYE (OPTIONAL — DARK ONLY)

“ON HARDCOAT, COLOR IS THE EXCEPTION, NOT THE RULE — AND ONLY DARK COLORS TAKE WELL.”

PURPOSE & THE “WHY”

If a color is specified (Class 2 dyed), draw **dark color down into the pores** of the fresh anodic oxide. **Most hardcoat is left its natural color** — a gray/bronze/dark that varies with the alloy — because the coating is thick, dense, and already dark, and its tight pores don’t take bright colors well. When hardcoat is dyed, it’s almost always **black** (or another dark color) to give a uniform dark finish over the variable natural color. After dyeing, the part **must be sealed** to lock the color in.



REMEMBER — DARK ONLY, AND DYE THEN SEAL

Bright reds/blues that pop on Type II come out **muddy or won’t take** on dense, dark hardcoat. Hardcoat dyeing means **black or dark shades**, and only when specified. If dyed, the order is still non-negotiable: **anodize → rinse → DYE → rinse → SEAL**.

PARAMETER	RANGE / PRACTICE	NOTES
Dye type	Organic black/dark (acid or basic); electrolytic for durable black	Dark shades only on hardcoat
Concentration	Per dye TDS	Higher/longer = deeper shade
Temperature	~50–60 °C typical	Affects uptake rate
pH	Per dye (often slightly acidic)	Critical for shade & consistency
Time	~10–20 min (thick coat, tight pores)	Until target dark shade
Water	DI	Tap-water impurities ruin color
Coating	Adequate thickness	Thickness helps darks; tight pores slow uptake



HEADS-UP — DYE HANDLING

Some dyes are **skin/eye irritants or sensitizers**, and dye baths may be warm and mildly acidic. Wear gloves and goggles, avoid skin contact, and read each dye’s SDS (some carry their own disposal requirements as colorants/heavy-metal content).



TIP — SET EXPECTATIONS ON HARDCOAT COLOR

Tell the customer (through your lead) that hardcoat’s **natural color varies by alloy** (6061 vs 7075 vs cast all look different), and that **only dark dyes** give a uniform result. Don’t promise a bright color on a hardcoat part — it won’t deliver. **Dye promptly** after the cool post-anodize rinse; the pores age and dye less well if the part sits.



TECHNICAL STUFF — WHY HARDCOAT RESISTS BRIGHT COLOR

Two reasons: the **coating’s own gray/bronze/dark natural color** shows through any light dye, and the **tight, dense pores** (the very thing that makes it hard) take up less dye than Type II’s open pores. Black “wins” because it overwhelms the natural color. For a very durable architectural-grade black, **two-step electrolytic coloring** (metal salts deep in the pores) is used — its own manual in this series.

STEP-BY-STEP

- Confirm dye type (dark), concentration, temp, pH (DI-based).
- Confirm part came from a cool DI rinse (pores open).
- Immerse fully, no trapped air; agitate gently to target dark shade.
- Drain, then **rinse** off surface dye.
- Move promptly to **seal**.

TOP MISTAKES

- **Sealing before dyeing** — fatal order error.
- Promising a bright color (won’t take).
- Inconsistent dwell/temp/pH → shade variation.
- Letting parts sit between anodize and dye.
- Hard/tap water in the dye line.
- Not rinsing surface dye → bleeding after seal.

SIGN-OFF — STATION 09

Teach-Back: Why hardcoat is usually natural or dark-dyed only; the correct order (anodize → rinse → dye → rinse → seal); why bright colors won’t take; the dye-handling hazards.

“I dyed (dark only) before sealing, into fresh pores, with consistent parameters in DI water, rinsed off surface dye, and moved promptly to seal — and I set realistic color expectations.”

Trainee: _____ Trainer: _____ Color/shade: _____

— STATION 10 OF 12

SEAL (OPTIONAL — THE WEAR-VS-CORROSION TRADEOFF)

“ON HARDCOAT, SEALING IS A CHOICE, NOT A HABIT — CLOSE THE PORES FOR CORROSION, OR LEAVE THEM OPEN FOR MAXIMUM WEAR.”

PURPOSE & THE “WHY”

Close the open pores of the anodic oxide so the coating reaches full corrosion resistance and any dye is locked in — but only when the part’s job calls for it. This is where hardcoat genuinely differs from Type II. On Type II you almost always seal. On hardcoat, sealing is a deliberate engineering tradeoff: it improves corrosion resistance (closes pores against salt/moisture and locks in dye) but slightly reduces hardness and abrasion resistance (the hydrated boehmite at the pore mouths is softer than the as-anodized oxide). So the print’s class tells you what to do:

- MIL-A-8625 Class 1 = unsealed — for maximum wear/hardness. Often not sealed, or instead impregnated with PTFE/dry-film lube.
- MIL-A-8625 Class 2 = sealed (and/or dyed) — for corrosion protection (and/or a uniform dyed color), accepting a small hardness penalty.

The classic method is hot DI water near boiling (boehmite swells the pores shut). There are also mid-temperature nickel-acetate and cold nickel-fluoride seals — each trading energy, speed, and performance, same as Type II.



REMEMBER — SEALING IS THE POINT OF NO RETURN, AND A REAL TRADEOFF ON HARDCOAT

After sealing you cannot dye (pores shut) and you slightly soften the wear surface. Follow the print: Class 1 = unsealed (max wear); Class 2 = sealed (corrosion/dyed). Don’t seal a Class 1 wear part out of habit — you’d throw away the hardness the customer paid for. And don’t skip sealing a Class 2 corrosion part. When unsure, ask — you can’t reverse either choice.

THE THREE COMMON SEALS

SEAL TYPE	TEMPERATURE	TRADEOFFS
Hot DI water seal	~96–100 °C	Simple, no metals, excellent corrosion; high energy, slow, prone to bloom; softens the wear surface most
Nickel-acetate (mid-temp)	~80–90 °C	Lower energy; excellent corrosion & dye retention; widely used; contains nickel
Cold nickel-fluoride	~25–30 °C	Lowest energy/fastest; often a warm “aging” rinse after; contains nickel + fluoride; tight control



TECHNICAL STUFF — WHY SEALING BOTH HELPS AND HURTS HARDCOAT

Near-boiling water hydrates the pore-wall oxide to boehmite (AlOOH), which swells and closes the pores — great for keeping corrosion out. But that swollen hydrated layer is mechanically softer than dense as-anodized Al₂O₃, so the top of a sealed hardcoat is a little less abrasion-resistant. On a wear part that small difference matters; on a corrosion part it’s a worthwhile trade — which is exactly why the spec sorts parts into Class 1 and Class 2.



TECHNICAL STUFF — “SEALING BLOOM” / SMUT

A powdery white/gray film after sealing (“bloom”) usually comes from hard water, drag-in, wrong pH, or over-concentrated seal depositing on the surface. It’s often removed with a light acid or proprietary post-dip — but the real fix is clean DI water and correct seal chemistry/pH/temperature.

**HEADS-UP — NEAR-BOILING TANK = SCALD HAZARD (IF SEALING HOT)**

A hot-water or hot nickel seal runs **~80–100 °C**. Splashes and steam cause **serious scald burns**. Use a face shield, heat-resistant gloves, long sleeves, and mind the steam cloud. Lower parts gently — a fast plunge throws hot water.

**HEADS-UP — NICKEL SEALS ARE A HEALTH & WASTE CONCERN**

Nickel is a skin sensitizer and the **fluoride** in cold seals is especially hazardous (deep, delayed burns; systemic toxicity). Gloves and goggles, read the SDS, know the fluoride first-aid (**calcium gluconate** per your plan), and route nickel/fluoride seal waste to proper treatment — **never to drain**.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Water	DI (hot-water seal)	Hard water → bloom
Temperature	Hot ~96–100 • Ni-acetate ~80–90 • Ni-F ~25–30 °C	Per chosen seal
pH	Per seal product (often ~5.5–6.5)	Wrong pH = poor seal/bloom
Time	~2–3 min per 0.0001 in (≈1 min/μm, hot water)	Thick hardcoat → long seal times
Result	Closed pores (Class 2); passes seal-quality test	Dye-stain or admittance test (Part Three)

**TIP — CLASS 1 MEANS DON'T SEAL**

If the traveler/print says **Class 1 (unsealed)** or calls for **PTFE impregnation**, the seal tank is **not** part of this part's route. Confirm before you dip — sealing a max-wear part is a quiet, unrecoverable spec violation.

STEP-BY-STEP

1. **Confirm the part's class** — Class 2 (seal) vs Class 1 (do **not** seal / PTFE instead). 2. Confirm seal type, temperature, pH, and DI water are in range. 3. Confirm the part came from a **clean rinse** after dye (if dyed). 4. Immerse gently (mind the scald), full time for the (thick) coating. 5. Withdraw, rinse if specified, inspect for **bloom**. 6. Move to final rinse/dry.

TEACH-BACK — CAN YOU ANSWER?

- What sealing does (closes pores) and the **tradeoff** (better corrosion, slightly less hard).
- Why Class 1 hardcoat is often left **unsealed**.
- Why you can't dye after sealing.
- The three seal types and one tradeoff of each.
- The scald hazard of hot seals; the nickel/fluoride hazard.

TOP MISTAKES

- **Sealing a Class 1 max-wear part** (throws away hardness).
- **Skipping seal on a Class 2 corrosion part.**
- Trying to dye after sealing (impossible).
- Wrong temp/pH → incompletely sealed.
- Hard/tap water → bloom.
- Routing nickel/fluoride waste to drain.

SIGN-OFF — STATION 10

"I confirmed the part's class first, sealed (Class 2) at the correct type/temperature/pH in DI water after dyeing — or correctly left it unsealed (Class 1) — checked for bloom, and handled the scald and (if applicable) nickel/fluoride hazards. I can explain the wear-vs-corrosion tradeoff."

Trainee: _____ Trainer: _____ Class / Seal type: _____

STATION 11 OF 12

RINSE / DRY

“FINISH CLEAN, DRY GENTLY, DON’T UNDO YOUR WORK.”

PURPOSE & THE “WHY”

Give the part a final clean rinse and a **gentle dry** without water spotting or thermal damage. A clean final rinse removes any drag-out so the part dries spot-free. Drying should be **gentle** — a finished hardcoat part is the end of a long, expensive process, and aggressive heat or hard-water spotting at the very end can blemish an otherwise perfect job.

PARAMETER	RANGE	NOTES
Final rinse water	DI	Spot-free drying
Rinse temperature	Warm	A warm final rinse aids drying
Dry method	Air / warm forced air; gentle oven	Avoid excessive heat
Handling	By rack / clean gloves	No bare-hand prints on finished parts

**HEADS-UP**
Parts coming off a hot seal are **hot** — burn risk. Let them cool or use gloves. Dry ovens are a heat hazard; mind hot racks.

**TIP**
A **DI final rinse** is the cheapest insurance against water spots on a finished part. Tap-water spots on a hardcoat surface are obvious and are pure rework on a high-value part.

STEP-BY-STEP

- Final DI rinse.
- Drain.
- Gentle dry.
- Handle by rack/gloves to inspection.

TOP MISTAKES

- Tap-water final rinse (spots).
- Over-aggressive drying.
- Bare-hand handling of finished parts.

SIGN-OFF — STATION 11

Teach-Back: Why a clean final rinse + gentle dry protects the finished job.

“I gave a clean DI final rinse, dried gently without spotting/overheating, and handled finished parts cleanly.”

Trainee: _____ Trainer: _____ Date: _____

STATION 12 OF 12

FINAL INSPECTION

“PROVE IT’S RIGHT — THICKNESS, DIMENSIONS, HARDNESS/WEAR, COLOR, AND (IF SEALED) SEAL.”

PURPOSE & THE “WHY”

Verify the finished part meets spec: **coating thickness**, **critical dimensions (after the large growth)**, **hardness/wear**, **color**, **overall appearance**, and — if **Class 2** — **seal quality**. Hardcoat quality isn’t visible by eye alone. **Thickness** (eddy-current; micro-section for thick coats) confirms you grew enough oxide; **dimensions** confirm the part is still in tolerance after the substantial growth; **hardness/abrasion** (microhardness or Taber, per spec) confirms it’s really *hardcoat* and not soft/burned; **seal quality** (only if sealed) confirms pores closed; and a visual check catches **burns**, bloom, cracking/crazing, pits, and bad contact marks.

CHECK	METHOD	PASS CRITERIA
Coating thickness	Eddy-current gauge; micro-section on significant surfaces	Within spec (e.g., ~50 µm / 0.002" per print)
Dimensions	Gauging on critical features	Within tolerance after dimensional growth
Hardness / wear	Microhardness and/or Taber abrasion (per spec)	Meets hardness/wear-index requirement
Seal quality (Class 2 only)	Dye-stain (ASTM B136) / admittance (ASTM B457)	Low dye pickup / low admittance = well sealed
Color	Visual vs. standard (natural or dark dye)	Matches approved sample
Appearance	Visual under good light	No burns , bloom, pits, cracking, bad contact marks



REMEMBER — THE CHECKS THAT MATTER MOST ON HARDCOAT

Thickness = “did we grow enough oxide?”, dimensions = “in tolerance after the big growth?”, and hardness/wear = “is it actually *hard*?” A hardcoat can be the right thickness and *still fail* if it’s soft (warm bath) or burned — which is why hardness/wear testing exists on this grade. Seal testing applies **only** to Class 2 sealed parts.



TIP — MICRO-SECTION FOR THICK COATS

Eddy-current is great for routine thickness, but on **thick hardcoat** (or for referee/disputes), a **microscopic cross-section** is the definitive measurement — it shows true thickness *and* reveals burning, cracking, or porosity that a gauge can’t.

STEP-BY-STEP

- Eddy-current (and micro-section as needed) thickness.
- Gauge critical dimensions (account for growth).
- Hardness/abrasion per spec.
- Seal test **if Class 2**.
- Color vs. standard; visual for burns/cracking.
- Accept / reject / route to rework (strip & re-anodize).

TOP MISTAKES

- Measuring thickness only on easy (not significant) surfaces.
- Missing dimensional growth on critical fits.
- Skipping hardness/wear verification.
- Running a seal test on a Class 1 (unsealed) part by mistake.
- Judging color under bad light.

SIGN-OFF — STATION 12

Teach-Back: Which check answers “enough oxide?” vs. “in tolerance?” vs. “hard enough?”; when the seal test applies (Class 2 only); why micro-section is used on thick coats.

“I verified thickness on significant surfaces, confirmed critical dimensions accounting for growth, verified hardness/wear, ran the seal test only where Class 2 applied, checked color, and inspected for burns/cracking.”

Trainee: _____ Trainer: _____ Thickness: _____ µm · Hardness: _____ · Seal (C2): P/F

PART THREE — ACROSS THE LINE
THE CROSS-CUTTING TOPICS

ACROSS THE LINE

THE IDEAS THAT DON'T LIVE IN ONE TANK — THEY TIE THE WHOLE LINE TOGETHER

• COATING THICKNESS & DIMENSIONAL GROWTH (DEEP DIVE)

Thickness is set by current density × time. At hardcoat's higher current density (~24–40 ASF) you grow oxide faster than Type II, but you run **longer** to reach the much greater thickness (~25–100+ μm). The same “minutes = (constant × mils) ÷ ASF” logic applies; your shop's exact constant comes from its own data and alloy. Always **verify with an eddy-current gauge** — and on thick coats or for referee, a **micro-section** — on the **significant surface**, not a convenient corner.

Dimensional growth is the operator-critical consequence, and it's **large** on hardcoat: oxide is bulkier than the metal, so the coating grows **~50% in, ~50% out**. Each surface moves out ~half the thickness; a **diameter changes by ~the full thickness** (pins grow, bores shrink). On a 50 μm (0.002”) coat that's ~0.001” per surface / ~0.002” on diameter; on 100 μm it doubles. Tight-tolerance, threaded, and press-fit features must be **machined to allow for growth or masked off** — and hardcoat drawings often specify **pre- and post-anodize dimensions separately**.



REMEMBER

Two numbers rule this section: **thickness = CD × time** (verify by eddy-current/micro-section) and **growth ≈ half-in/half-out** (and on hardcoat that's *big* — plan tolerances and masking around it).

• RACKING, ELECTRICAL CONTACT & HEAT (DEEP DIVE)

The part is the anode, so **the rack is the wire that grows your coating** — and on hardcoat that wire carries **much more current**. Good hardcoat racking means: **generous, firm contacts** (more and bigger than Type II to avoid contact-area burning); contact **placed where the bare mark won't matter**; **orientation for gas escape and drainage**; **no parts touching**; **strong agitation reaching all surfaces** (to carry away the heat the high power generates); and **maintained racks** (titanium preferred; aluminum racks stripped/re-pointed as they hardcoat over). Poor contact and poor heat removal are the two most common causes of **burning, thin coatings, and dropped parts**. (*Cross-reference Station 01.*)

• COOLING & “BURNING” (DEEP DIVE)

The deep dive unique to hardcoat. The process dumps a lot of electrical energy into the part, and **cold suppresses oxide dissolution** — which is what makes the coating hard. So two things must always be true: the bath must be held **near 0 °C** by a **powerful refrigeration system**, and **strong agitation** must carry heat away from the part surface. If either falters, the part **burns** (local overheating → powdery/charred oxide, sometimes melted/pitted metal — scrapped). Defenses: **aggressive cooling, strong agitation, a controlled voltage ramp, generous contacts, a current density within the alloy's limit, and never set-and-forget**. Burning is *the* hardcoat failure mode — and a safety as well as a quality issue (runaway heat in an acid tank).

• DYEING & SEALING — THE WEAR-VS-CORROSION TRADEOFF (DEEP DIVE)

On hardcoat, **dye is usually skipped** (natural gray/bronze) or limited to **dark colors (black)**, because the dense, dark, tight-pore coating won't take bright shades. **Seal is a deliberate choice, not a habit**: sealing **improves corrosion resistance but slightly reduces hardness/abrasion** (boehmite is softer than as-anodized oxide). The print's **class** decides: **Class 1 = unsealed** (max wear; sometimes PTFE-impregnated instead), **Class 2 = sealed** (corrosion/dyed). Seal routes are the same three as Type II — **hot DI water (~96–100 °C)**, **nickel-acetate (~80–90 °C)**, **cold nickel-fluoride (~25–30 °C)** — all enemies of **hard water, wrong pH, and contamination** (which cause **bloom**). If sealed, confirm with a **dye-stain (ASTM B136)** or **admittance (ASTM B457)** test.



REMEMBER

Dye = dark color (or none). Seal = a tradeoff: corrosion up, hardness slightly down. Follow the class — Class 1 unsealed for wear, Class 2 sealed for corrosion. Test the seal *only* when the part is sealed.

• ALUMINUM ALLOY EFFECTS (DEEP DIVE)

The alloy decides how hard, how thick, what color, and how *risky* the hardcoat is. **Wrought 6061/6063/7075** hardcoat **hard, clean, and predictable** (natural gray/bronze). **Copper (2xxx, e.g., 2024)** is **difficult** — lower hardness, darker/uneven, and a high **burn risk**, often needing a gentler ramp, lower CD, colder bath, and reduced thickness. **Silicon (cast alloys)** turns the coating **dark/charcoal, lower in hardness**, and mottled. Rack **like alloy with like alloy**, and run a **sample of the actual alloy/lot** before promising hardness/thickness/color on a tough alloy. (Cross-reference Chapter 6.)

• RINSING & WATER/DI MANAGEMENT

Rinses are firewalls between opposite chemistries (caustic etch vs. acid desmut/anodize) and protectors of the (cold, sensitive) anodize bath and any dye/seal. **Counter-current cascade** rinsing saves water; **conductivity** is the cleanliness gauge (lower = cleaner). Use **DI water** for the post-anodize rinse, dye, seal, and final rinse — tap-water impurities (especially **chloride** and hardness) cause pitting, off-color dye, and sealing bloom.

• WASTE & ENVIRONMENTAL

A hardcoat line generates several regulated streams — **never to a floor drain untreated**:

- **Spent sulfuric acid & acid rinses** — neutralize/treat per permit; manage **dissolved aluminum** (precipitated as aluminum hydroxide sludge). *(Hardcoat baths build dissolved Al and need periodic decant/treatment.)*
- **Spent caustic etch** — high pH; contains dissolved aluminum; neutralize/treat.
- **Desmut waste** — acidic; **fluoride-bearing** desmuts need fluoride treatment (precipitate as calcium fluoride).
- **Dye waste** (if dyed) — colorants; some contain regulated metals; treat/dispose per SDS and permit.
- **Nickel-seal waste** (if sealing with nickel-acetate/nickel-fluoride) — **nickel is regulated**; fluoride needs treatment; segregate and treat — never to sewer.
- **Air** — acid mist (sulfuric) and steam need exhaust/scrubbing per permit.



HEADS-UP

Acidic and caustic streams must be **neutralized under control** (never casually mixed), and nickel/fluoride streams need **dedicated treatment**. Know your shop’s segregation scheme; the wrong drum or drain can be a reportable violation.

• QUALITY CHECKS & TEST METHODS

CHECK	METHOD (TYPICAL STD)	ANSWERS
Coating thickness	Eddy-current (ASTM B244); micro-section for thick coats	“Enough oxide grown?”
Hardness / wear	Microhardness ; Taber abrasion (per MIL-A-8625 / AMS 2469)	“Hard/wear-resistant enough?”
Dimensions	Gauging on critical features	“In tolerance after growth?”
Seal quality (Class 2)	Dye-stain (ASTM B136); admittance (ASTM B457)	“Pores closed?”
Corrosion (esp. Class 2)	Salt spray (ASTM B117) per spec	“Will it survive service?”
Color	Visual / spectrophotometer vs. standard	“Right natural/dark shade?”
Appearance/defects	Visual under controlled light	Burns, cracking, bloom, pits, marks

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Burning (powdery, dark, charred — esp. edges/contacts)	Cooling lost; CD too high; ramped too fast; weak agitation; undersized contacts	Restore cooling/agitation; lower CD; slow ramp; add/bigger contacts; verify ~0 °C
Soft / low hardness	Bath too warm; high dissolved Al; (over-)sealing	Restore cold temp; lab-check Al; reconsider seal per class
Thin or no coating	Poor contact; wrong polarity (part not anode) ; low current; low acid; stalled ramp	Re-rack tight; verify part = anode ; check current/acid/ramp
Pitting	Chloride contamination ; impurities	Lab-check chloride; switch to DI; treat bath
Thickness/hardness varies part-to-part	Mixed alloys; inconsistent temp/CD/dwell	Rack like-with-like; standardize ramp/timing; hold temp
Dull / dark / mottled color	High-Si or high-Cu alloy (inherent); high dissolved Al; smut left on	Confirm alloy; lab-check Al; verify desmut
Cracking / crazing of thick coat	Very high thickness + thermal stress (can be inherent)	Confirm thickness vs spec; discuss with chemist
Won't take dark dye / blotchy	Coating too thin or too tight; weak/old dye; pre-sealed	Confirm thickness; refresh dye; don't pre-seal before dye
Poor seal (Class 2: fades/stains/fails test)	Wrong seal temp/pH; contaminated seal; pre-sealed before dye	Correct temp/pH/DI; re-seal; don't pre-seal
Sealing bloom (chalky film)	Hard water; wrong pH; over-concentrated seal	DI water; correct pH/concentration; light post-dip
Contact burns / no-coat at contact	Loose/undersized rack contact; current crowding	Add/enlarge contacts; re-point/replace rack
Sealed a Class 1 part by mistake	Wrong route / didn't check class	Verify class before seal; strip & re-anodize if wrong



TIP

Most hardcoat defects trace to one of four roots: **temperature/cooling** (burning, soft coat — *the* hardcoat killer), **power/ramp & contacts** (burning, thin coat), **contamination** (chloride/hard water — pitting & bloom), or **alloy** (dark/soft/burn-prone). Name the root, not just the symptom.



REMEMBER

Several of these are **safety-relevant** as well as quality: **wrong polarity** (verify the part is the anode), **cooling loss / burning** (runaway heat in an acid tank), and any move that increases acid attack or current. When in doubt about a bath or power change, **ask your supervisor before you act**.

— EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

PLAIN-LANGUAGE DEFINITIONS FOR EVERYTHING IN THIS MANUAL

Admittance test: an electrical test (ASTM B457) of how “open” the coating still is — confirms sealing (lower = better seal). *Applies to sealed (Class 2) parts.*

AMS 2469: an aerospace specification for hardcoat (Type III) anodizing.

Anode: the **positive** electrode. In anodizing it's **your part** — where the oxide grows.

Anodizing: growing an integral oxide layer out of a metal's surface by making the part the anode in an electrolyte.

Barrier layer: the thin, dense oxide at the bottom of the coating, against the metal (under the porous layer).

Boehmite: hydrated aluminum oxide (AlOOH) formed during hot-water sealing; its swelling closes the pores (and slightly softens the surface).

Burning: local overheating during anodizing that turns the coating powdery/charred (and can melt/pit the part) — *the* hardcoat failure mode.

Cathode: the **negative** electrode (lead or aluminum here) — carries current and bubbles off hydrogen; does **not** feed the coating.

Chloride contamination: dissolved chloride in the bath that causes **pitting/burning**; kept very low (often $\leq \sim 50$ ppm).

Class 1 / Class 2 (MIL-A-8625): **Class 1 = unsealed** hardcoat (max wear); **Class 2 = sealed (and/or dyed)** (corrosion/color).

Current density (ASF / A/dm²): amps per unit area. Hardcoat $\approx 24\text{--}40$ ASF ($\sim 2.5\text{--}4$ A/dm²) — higher than Type II.

Desmut / deox: an acid dip that removes the dark **smut** left by caustic etching.

Dimensional growth: the coating grows **~half in, half out**; each surface moves out ~half the thickness.

Drag-out / drag-in: solution clinging to a part leaving a tank (out) / contaminating the next tank (in).

Dye: colorant absorbed into the pores — on hardcoat, **dark colors only**, and optional.

Eddy-current gauge: non-destructive instrument for measuring coating thickness on aluminum.

Etch: a caustic dip that removes the natural oxide and a little aluminum — run **lighter** on hardcoat to preserve dimension.

Hardcoat (hard anodize / Type III): thick, dense, very hard sulfuric anodize run **cold and at high power** for wear/abrasion resistance.

Integral coating: a coating that *is* part of the metal (grown from it), not added on top — can't peel to bare metal.

Microhardness: a measured hardness of the coating (e.g., Knoop/Vickers) — used to verify a hardcoat is actually hard.

MIL-PRF-8625 (MIL-A-8625): the spec defining anodize **Type I (chromic)**, **II (sulfuric)**, **III (hardcoat)** and classes.

Porous layer: the upper oxide full of vertical **pores** — on hardcoat, tight and dense (which is why it's hard).

Ramp (voltage ramp): gradually raising the voltage through a hardcoat run to avoid current spikes and burning.

Sealing: an optional step that **closes the pores** — on hardcoat a **tradeoff** (better corrosion, slightly less hard).

Sealing bloom (seal smut): a chalky film after sealing, usually from hard water / wrong pH / contamination.

Significant surface: the surface that must meet thickness/hardness/finish spec; where you measure.

Smut: dark residue of alloying elements (Cu, Si, Fe) left on the surface after caustic etching.

Taber abrasion: a wear test (per spec) used to confirm hardcoat's abrasion resistance.

Type III: the **hardcoat** anodize — thick, dense, hard, wear-resistant; sulfuric, cold, high-power.

Valve metals: metals that anodize to form protective oxides (aluminum, titanium, magnesium, tantalum, niobium).

Water-break test: a free cleanliness check — clean metal sheets water; dirty metal beads it.

★ REMEMBER

Of all these, five matter most for hardcoat: **anode** (your part), **cold-makes-hard** (the whole point of Type III), **burning** (the failure mode — watch the cooling), **big dimensional growth** (plan your tolerances), and the **seal-vs-wear tradeoff** (Class 1 unsealed vs Class 2 sealed). Master those and you understand Type III.

☀ FUN FACT

The word “**anode**” comes from the Greek *ánodos*, “a way up” (Faraday's term for the electrode where current enters the cell). On this line, that “way up” is literally where your thick, hard coating grows — a fitting name for the star of the show.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Safety / PPE / SDS / hygiene — acid, caustic, cold bath, hot seal, hydrogen, high-power DC (all stations)	_____	_____	_____	_____
2	Station 01 — Inspection & Racking (high-current contact)	_____	_____	_____	_____
3	Station 02 — Clean / Degrease + water-break	_____	_____	_____	_____
4	Station 03 — Etch (caustic, light for hardcoat)	_____	_____	_____	_____
5	Station 04 — Rinse + conductivity	_____	_____	_____	_____
6	Station 05 — Desmut / Deox	_____	_____	_____	_____
7	Station 06 — Rinse	_____	_____	_____	_____
8	Station 07 — Anodize (part = anode; cold; voltage ramp)	_____	_____	_____	_____
9	Cooling / burning control & temperature hold	_____	_____	_____	_____
10	Thickness control (CD × time) & large dimensional growth	_____	_____	_____	_____
11	Station 08 — Post-anodize rinse (fresh coating)	_____	_____	_____	_____
12	Station 09 — Dye (dark only, optional)	_____	_____	_____	_____
13	Station 10 — Seal (Class 1 vs Class 2 / wear-vs-corrosion)	_____	_____	_____	_____
14	Station 11 — Rinse / Dry	_____	_____	_____	_____
15	Station 12 — Final Inspection (thickness/dims/hardness)	_____	_____	_____	_____
16	LOTO on rectifier / high-power electrical safety	_____	_____	_____	_____
17	Emergency response — eyewash, shower, spill, fluoride first-aid	_____	_____	_____	_____
18	Waste segregation & environmental controls	_____	_____	_____	_____
19	Troubleshooting — symptom recognition (esp. burning)	_____	_____	_____	_____

Operator name: _____ Hire/start date: _____

Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____



HEADS-UP

Safety rows (1, 9, 16, 17, 18) are not optional and not “fill in later.” An operator may not work any station until PPE/SDS/hygiene, cooling/burning control, LOTO/high-power electrical, emergency response (including fluoride first-aid where used), and waste-control awareness are signed. Safety competency precedes production competency.

PART FOUR — CERTIFICATION

20 QUESTIONS • PASSING SCORE 80% (16/20)

COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

**REMEMBER**

Passing score: 80% (16 of 20). Any **safety** question answered wrong (**Q3, Q9, Q13, Q17, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs: 3, 9, 13, 17, 19**

Q1. Anodizing differs from electroplating mainly because in anodizing the part is the:

- A. Cathode, and a different metal is deposited on it B. Anode, and an oxide is grown out of the part itself C. Insulator, and nothing happens D. Cathode, and it dissolves

Q2. The Type III hardcoat electrolyte is:

- A. Chromic acid B. Sodium hydroxide C. Dilute sulfuric acid (same family as Type II, but run cold) D. Nickel sulfate

Q3. [SAFETY GATE] The single most important reason the hardcoat bath must be held **near 0 °C** is:

- A. To save electricity B. Cold suppresses oxide dissolution so the coating grows dense and hard — and the high-power process makes a lot of heat that, uncontrolled, *burns* the part (a hazard) C. To freeze the part D. Temperature doesn't matter

Q4. Typical Type III coating thickness is about:

- A. 0.5–1 µm B. ~25–100+ µm (commonly ~50 µm / 0.002") C. 1–2 inches D. There is no coating

Q5. Compared to Type II, hardcoat (Type III) is run with:

- A. Warmer bath, lower voltage B. Colder bath, higher voltage and current density C. No electricity D. The same conditions exactly

Q6. (Short answer) Explain in your own words why anodizing is **not** plating — name which electrode the part is and **where the coating comes from**.

Q7. "Dimensional growth" on hardcoat means the coating grows roughly:

- A. All outward B. All inward C. Half into the surface and half out — and because hardcoat is thick, this is a large, tolerance-critical number D. It shrinks the part

Q8. The wear-vs-corrosion **sealing tradeoff** on hardcoat is:

- A. Sealing always improves everything B. Sealing improves corrosion resistance but slightly reduces hardness/abrasion — so max-wear (Class 1) parts are often left unsealed C. Sealing makes the part harder D. You can never seal hardcoat

Q9. (Short answer) [SAFETY GATE] Name the **two corrosive hazards** on the line (one high-pH, one low-pH), and state the universal rule for adding acid during bath make-up.

Q10. Typical Type III voltage and current density are about:

- A. 1 V; 0.1 ASF B. ~25–60+ V (ramped); ~24–40 ASF C. 200 V; 500 ASF D. 1000 V; 1 ASF

Q11. A hardcoat part comes out **dark, mottled, and soft** even though the line is running. The most likely cause is:

- A. The bath is too clean B. A high-silicon (cast) or high-copper (2xxx) alloy, which hardcoat poorly C. Too much dye D. The rectifier is off

Q12. (Short answer) What is “burning” on a hardcoat line, what causes it, and name two ways to prevent it.

Q13. [SAFETY GATE] Compared to Type II, the hardcoat rectifier runs at **higher voltage and current**. The main added hazard is:

A. Radiation B. Greater electrical shock / arc hazard at contacts and bus (LOTO, dry hands) C. None D. Magnetism

Q14. Why can't you **dye** a hardcoat part after it has been **sealed**?

A. The dye is too expensive B. Sealing closes the pores, so there's nowhere for the dye to go C. You can — order doesn't matter D. The part is too cold

Q15. Coating **thickness** in hardcoat is controlled mainly by:

A. Dye color B. Current density × time C. Seal temperature D. The alloy only

Q16. (Short answer) What do **MIL-A-8625 Class 1** and **Class 2** mean for hardcoat, and which one is chosen for maximum wear?

Q17. (Short answer) [SAFETY GATE] List **three** pieces of PPE required at the sulfuric hardcoat/desmut tanks, and name the special first-aid concern if your desmut or cold seal contains **fluoride**.

Q18. Chloride contamination in the hardcoat bath causes:

A. Brighter color B. Pitting / burning of the coating C. Faster sealing D. Nothing

Q19. (Short answer) [SAFETY GATE] Name the **flammable gas** produced at the cathode during anodizing and **two** controls that keep it from becoming a fire hazard.

Q20. (Short answer) Name **two** ways hardcoat (Type III) differs from standard Type II anodizing.



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 3, 9, 13, 17, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

**HEADS-UP**

Keep this page out of the trainee's copy. Questions **3, 9, 13, 17, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Q1. B — The part is the **anode**, and oxide is grown out of the part itself.

Q2. C — Dilute sulfuric acid (same family as Type II), but run **cold and at higher power**.

Q3. B — **Cold suppresses oxide dissolution** so the coating grows **dense and hard**; and the high-power process makes a lot of heat that, if cooling fails, **burns** the part — a quality *and* safety hazard. (Held near 0 °C with strong refrigeration + agitation.)

Q4. B — ~25–100+ µm (commonly ~50 µm / 0.002").

Q5. B — **Colder bath, higher voltage (ramped), and higher current density** than Type II.

Q6. In plating the part is the **cathode** and a **different metal is deposited onto it** from the bath. In anodizing the part is the **anode**, and the coating is **aluminum oxide grown out of the part's own surface** (oxygen from the bath combines with the part's aluminum) — nothing is deposited from elsewhere.

Q7. C — Half in, half out; each surface grows out ~half the thickness (a diameter changes by ~the full thickness) — and on thick hardcoat that's a **large, tolerance-critical** number.

Q8. B — Sealing **improves corrosion resistance but slightly reduces hardness/abrasion**, so max-wear **Class 1** parts are often left **unsealed**; **Class 2** parts are sealed for corrosion (and/or dyed).

Q9. The **caustic etch** (high pH / sodium hydroxide) and the **sulfuric anodize/desmut** (low pH / acid) — both cause severe burns (cold acid included). Universal rule: **always add ACID to WATER, never water to acid**.

Q10. B — ~25–60+ V (ramped; can reach ~75 V on thick coats); ~24–40 ASF (~2.5–4 A/dm²).

Q11. B — A **high-silicon cast alloy** or a **high-copper 2xxx alloy** — both hardcoat poorly (dark, mottled, lower hardness, burn-prone). The alloy decides how good the hardcoat can be.

Q12. Burning is local runaway overheating during anodizing that turns the coating **powdery/charred** (and can melt/pit the part), often at edges or contacts. Causes: **lost cooling, too-high current density, ramping too fast, weak agitation, undersized contacts**. Prevention (any two): **aggressive cooling / hold ~0 °C; strong agitation; controlled voltage ramp; generous contacts; current density within the alloy's limit**.

Q13. B — Greater **electrical shock / arc hazard** (hardcoat runs more volts and amps). Use LOTO and dry hands; respect the bus and contacts.

Q14. B — Sealing **closes the pores**, so dye has nowhere to go. Dye must come *before* seal.

Q15. B — Current density × time (verify with an eddy-current gauge / micro-section).

Q16. Class 1 = unsealed hardcoat (chosen for **maximum wear/hardness**); **Class 2 = sealed (and/or dyed)** hardcoat (chosen for corrosion protection and/or color). **Class 1 is for maximum wear**.

Q17. PPE (any three): sealed chemical splash goggles, face shield, chemical-resistant gloves, apron/suit, chemical-resistant boots. **Fluoride concern:** fluoride causes **deep, delayed, systemically toxic burns** — requires specific first-aid (e.g., **calcium gluconate** per SDS/medical plan) and immediate medical attention; ordinary flushing alone may not be enough.

Q18. B — Pitting/burning of the coating. Keep chloride very low (use DI water).

Q19. Hydrogen gas (at the cathode). Controls (any two): **ventilation/exhaust running, no open flames/sparks/grinding nearby**, no smoking, keep ignition sources away.

Q20. Any two of: hardcoat is run **colder (near 0 °C)**; at **higher voltage (ramped) and current density**; is **thicker (~25–100+ µm)**; is **much harder / more wear-resistant**; has **larger dimensional growth**; is usually **natural or dark-dyed only**; uses **Class 1 unsealed vs Class 2 sealed** (the seal-vs-wear tradeoff); and carries a greater **burning** risk requiring more cooling.

**REMEMBER**

A score of 16/20 passes — but every safety question (3, 9, 13, 17, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people.

— PRINT, FILL IN, SIGN



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CERTIFICATE OF
COMPLETION

HARDCOAT ANODIZING (TYPE III) TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Hardcoat Anodizing (Type III)** training course — covering the full hardcoat workflow (Inspection & Racking, Clean, Etch, Rinse, Desmut/Deox, Anodize, Dye, Seal, Rinse/Dry, and Final Inspection), the fundamental concept that **the part is the anode and the oxide is grown out of the part itself**, the **cold-makes-hard** principle and **ramped high-power** operation, the **seal-vs-wear tradeoff (Class 1 unsealed vs Class 2 sealed)**, the **large dimensional growth** of a thick coating, coating-thickness control by current density and time, high-current electrical-contact and racking practice, aluminum-alloy effects and burn risk, **cooling/burning control**, hardness/wear verification, and — above all — the **safety practices** required of every operator (sulfuric acid and caustic-etch burn control, cold-bath and refrigeration awareness, **higher-voltage DC** and arc controls, hydrogen controls, hot-seal scald control where applicable, fluoride first-aid where applicable, spill response, and waste segregation) — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer/aerospace specifications (e.g., MIL-PRF-8625 Type III, AMS 2469), and applicable regulatory requirements before production use. Sulfuric acid and caustic etch cause severe burns; the hardcoat bath runs near freezing under high-current DC, and seal tanks run near boiling — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

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